



សាកលវិទ្យាល័យជ្យាយង
LIFE UNIVERSITY

ប្រព័ន្ធក្រុឌក្នុងកសាសនាឌីជីថលនៅ
សាកលវិទ្យាល័យជ្យាយង

A Digital Thesis Repository Platform at Life University

Research Thesis Presented to
The College of Science and Engineering Department of Computer Science
Life University, Preah Sihanouk, Cambodia

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Academic Year (2023-2026)

សាកលវិទ្យាល័យ ឡាយហ្វ

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KINGDOM OF CAMBODIA

Nation Religion King

LIFE UNIVERSITY

College of Science and Engineering

**SUPERVISORY COMMITTEE APPROVAL FACTORS AFFECTING
STUDENT'S COMPUTER SCIENCE ACHIEVEMENT:
A Digital Thesis Repository Platform at Life University**

Prepared and submitted by Phai LyChhy, Chhin SreyPich, and Ol Poleak in partial fulfillment of the requirements for the degree of Bachelor of Computer Science, has examined and is recommended for acceptance and approval for a passing grade by the supervisory committee.

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ABSTRACT

This project presents the development of a Digital Thesis Repository Platform at Life University to improve the storage, management, search, and accessibility of thesis documents. The system was developed to address the limitations of physical and scattered thesis storage, which can make research materials difficult to manage and access. The platform provides a centralized repository for completed and passed theses and includes features such as thesis search and filtering, PDF viewing and downloading, thesis publication requests, approval and rejection, notifications, view history, bookmarks, and email verification. Role-based access control was implemented for Administrators, Head of Departments, Students and Guests to ensure that users can access functions according to their responsibilities. The system was developed using the Software Development Life Cycle (SDLC) with the Waterfall model. The technologies used include PHP, Laravel, MySQL, HTML, CSS and JavaScript. System testing was conducted using the Black-Box Testing approaching, covering functional testing, role-based access testing, and email verification testing. A total of 42 test cases were conducted, and all 42 cases passed, which gives a 100% pass rate. The system performance testing using k6 also showed that the system successfully handled a combined workload of 130 virtual users. The user evaluation achieved a 74.26% positive response rate, indicating overall user satisfaction with the system. Overall, the developed system provides a centralized and organized platform for managing and accessing Life University theses and supports more convenient access to academic research materials.

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List of Abbreviation

AI	:	Artificial Intelligence
CSS	:	Cascading Style Sheets
Dept	:	Department
DFD	:	Data Flow Diagram
DR	:	Digital Repository
DTR	:	Digital Thesis Repository
DTRP	:	Digital Thesis Repository Platform
DV	:	Dependence Variable
EDR	:	Entity Relationship Diagram
ETD	:	Electronic Theses and Dissertations
HEI	:	Higher Education Institution
HTML	:	Hyper Text Markup Language
HTTP	:	Hypertext Transfer Protocol
HoD	:	Head of Department
HVD	:	Harvard University Digital Repository
IR	:	Institutional Repository
IV	:	Independent Variable
JS	:	Javascript
LU	:	Life University
MoEYS	:	Ministry of Education, Youth and Sport
MySQL	:	My Structured Query Languages
NUM	:	National University of Management

OAR	:	Open Access Repository
OPR	:	Overall Positive of Agree
PDF	:	Portable Document Format
PHP	:	Hypertext Preprocessor
SDLC	:	Software Development Life Cycle
SPA	:	Sum of Percentage of Agree
SPAS	:	Sum of Percentage of Strongly Agree

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Universities around the world produce a large amount of academic research through student theses, reports, and dissertations every year. These academic works provide many important sources of knowledge, ideas, examples, and innovation within their respective fields and allow new researchers to learn from previous research studies. These works also serve as valuable academic resources for researchers and future students who seek access to relevant academic information for their studies. The proper management and preservations of these works are essential to ensure long-term accessibility, reduce the risk of loss, and share the research findings with others, which improves the quality of learning and study at the university.

Traditionally, print-based thesis management system face limitations in supporting the growing volume of graduate research. Each year, thousands of theses and dissertations are produced continuously, leading to an expansion of printed collections, while the increasing number of physical documents creates challenges related to storage space, preservation, and long-term maintenance within university libraries (Edward A. Fox et al., 1996). Furthermore, printed thesis documents are typically accessible only in the library, which makes it harder to share research and restrict access for readers. These limitations indicate that print-based systems lack effective management of research outputs.

In 1997, a transformation occurred when many American universities began to require Electronic Theses and Dissertations (ETDs) submissions to improve access and equity for researcher (Wiederhold & Nimer, 2023). Virginia Tech became the first universities to require electronic submission of all graduate theses and dissertations on

January 1, 1997. Since then, Virginia Tech graduate students have been able to prepare, submit, review, and publish their theses and dissertations online. The University libraries have converted thousands of older printed theses and dissertations into digital format and made all of the documents available in VTechWorks digital repository, providing public access to these documents (Virginia Tech, n.d.).

In many developing countries, including Cambodia, higher education institutions (HEIs) are slowly moving toward digital transformation to improve teaching, learning, and research outcomes (Edman Flores, Udam Mean, 2024). Despite this progress, the management of student theses remains traditional in some universities where theses are stored in printed form or separate digital file in departmental or library storage. According to Sreyphos (2025), the Ministry of Education, Youth and Sport (MoEYS) has encouraged collecting and sharing student theses in national research archives and has called for public access to support future scholars. At Life University (LU), in Sihanoukville, Cambodia, the management and accessibility of completed theses present an opportunity for digital improvement. A centralized digital repository can help address these challenges by providing a structured platform for storing and accessing approved theses.

Therefore, this study proposes the development of **A Digital Thesis Repository Platform in Life University**. The system is designed to allow students whose final theses have passed the required academic evaluation to request publication of their theses in the digital repository. Authorized Heads of Department (HoDs) or Administrators can review the publication requests and approve or reject them before the theses are added to the repository. Once approved, the thesis can be organized and made searchable according to relevant such as title, author, department, degree level, and year.

The development of this platform is expected to improve the organization and

accessibility of completed theses at Life University. It can also reduce physical document storage and provide students and researchers with a convenient way to discover previous academic research.

1.2 Statement of the Problem

The current management and accessibility of completed theses at Life University (LU) face several challenges related to physical storage, accessibility, and thesis management. Based on the current situation, the following problems have been identified:

a. Storage and Preservation

- Physical theses require storage space and are difficult to organize and maintain
- Physical documents are at risk of loss or damage due to handling and borrowing.
- Completed theses need digital preservation for long-term access.

2. Accessibility

- Students may have difficulty finding and accessing previous theses for reference.
- Access is limited by library operating hours and thesis availability.
- Students need to visit the physical library or contact departmental staff to access a thesis
- Students need to wait for another student to return a borrowed thesis before accessing it.
- Searching, borrowing, and printing theses can take time and cost money.

3. Management and Publication

- There is no centralized digital repository for storing and accessing

approved theses.

- Manual management of thesis records and borrowing can be inconvenient and time-consuming.
- A structured process is needed for students who have already passed their theses to request publication, and for HoDs or Administrators to approve or reject the requests.

These problems indicate the need for a centralized digital thesis repository that can improve the storage, accessibility, preservation, and management of completed theses at Life University.

1.3 Objective of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a Digital Repository Platform (DTRP) in Life University (LU) to improve the management of theses storages, and accessible for students, lecturers, and researchers, both inside and outside the university.

1.3.2 Specific Objectives

This study aims to transform the existing manual thesis management system process into a digital platform that can improve the storage, management, and accessibility of completed theses. Specifically, the study aims:

- To design and develop a centralized digital repository for storing and preserving completed theses.
- To develop a process that allows students who have passed their final theses to request publication in the repository and authorized HoDs or Administrators to review and approve or reject the requests.

- To provide search and filtering functions to help users locate relevant theses efficiently.
- To provide secure online access to approved thesis documents.
- To organize thesis information such as title, department, and year.
- To reduce the time, cost, and use of physical documents when managing and accessing theses.
- To evaluate the usability and performance of the developed platform.

1.4 Research Questions

This study is guided by the following research questions:

1. What challenges do students face when accessing and obtaining previous thesis document at Life University?
2. How do students currently search for and obtain previous thesis documents for study and reference?
3. What difficulties do students experience in terms of time, cost and accessibility when obtaining previous thesis document?
4. What features and functions do students need a digital thesis repository to make accessing, searching, and obtaining previous thesis documents easier and more efficient?

1.5 Scope and Limitations of the Study

This study focuses on the design and development of a web-based Digital Thesis Repository Platform (DTRP) for Life University (LU). The system is intended to support the management, publication, organization, preservation, and accessibility of final theses that have already passed the required academic evaluation. The scope and limitations of the study are as follows:

1. User Management

Admin:

- Create and manage Head of Department (HoD) accounts.
- Enter and manage HoD information, including full name, department, email, and password.
- Activate or terminate HoD accounts when necessary.
- Create and manage departments.
- Manage thesis documents from all departments.
- Review, approve, or reject students' publication requests and provide feedback when requests are rejected.
- Browse, search, filter, view, and download approved thesis documents.
- The Admin cannot create a thesis publication request as a student does.

Head of Department (HoD):

- Log in using an account created by Admin
- Change their username
- Manage thesis documents within their own department.
- Review, approve, or reject students' thesis publication requests within their department and provide feedback when requests are rejected.
- Browse, search, filter, view and download approved thesis documents from all departments.
- The HoD cannot create thesis publication requests a student does.

Student:

- Log in using a school email address and verify the email.
- Manage their personal account information.
- Browse, search, and filter approved theses by relevant information such as title, department, and year.
- View and download approved thesis documents.
- Request publication of a final thesis that has already passed the required academic evaluation
- Only a fourth-year student who is the team leader of a thesis group can request publication of the group's final thesis.
- Provide the required personal and thesis information to obtain permission to request publication.
- Wait for the Admin or HoD to review the publication request.
- Resubmit a request after rejection when the required issues have been corrected.

Guest User or Researcher:

- Log in with a general Email Address.
- Manage personal account information.
- Browse, search, and filter theses by relevant information such as title, department and year.
- View and download thesis documents.

2. System Management

- The Admin has the highest level of access and can manage major system functions and HoD accounts.
- HoDs have access to thesis management and publication approval

functions within their assigned departments.

- Students do not have upload permission by default.
- Upload and publication permission is provided only to eligible fourth-year students who are team leaders of their thesis groups.
- A thesis must have already passed the required academic evaluation before it can be requested for publication.
- Thesis publication requires approval from an authorized Admin or HoD.
- Admins and HoDs review publication requests to prevent unauthorized or inappropriate thesis documents from being added to the repository.
- The system stores and manages thesis documents in PDF format only.
- Approved theses can be organized and searched using information such as title, author, department, and year.
- Approved thesis documents can be viewed and downloaded through the platform.

3. Limitations of the Study

- The system focuses only on the management, publication, organization, preservation, and access of final theses that have already passed the required academic evaluation.
- The system does not manage the thesis writing or research development process.
- Thesis defense and academic evaluation are outside the scope of the system.

- Thesis grading are not included.
- Plagiarism checking is not included.
- Advanced Artificial Intelligence (AI) features are not included.
- The system does not manage other types of academic documents such as assignments, reports, journal articles, or dissertations.
- Thesis documents are limited to PDF format.
- The system does not determine whether a thesis has passed academically.

1.6 Significance of Study

The significance of this study is the proposed DTRP within LU aim will have benefits for different stakeholders:

1. Students: Easily search and access previous theses and request to publish their final theses in the repository.
2. Lecturers and Researchers: Access previous theses for teaching, research, and academic reference.
3. Heads of Departments (HoDs): Review, approve, reject, and manage thesis publishing requests within their departments.
4. Administrators: Manage users, departments, publication requests, and theses across the university.
5. Life University: Improve the storage, organization, preservation, and accessibility of academic research.
6. Future Researchers: Use previous theses as references for future studies.

For students, the system provides a centralized location to easily view, browse, search and download previous theses for academic learning. This allows students in early years to understand thesis topics, structures, and research methodologies by learning from

the example of previous graduates and reduces the time and effort without requiring visits to the physical library for access, especially when access is limited due to the library's operation hours.

1.7 Definitions of Key Terms

The following are explanations of key terms used in this study:

1. Digital Thesis Repository Platform (DTRP): Refers to the web-based system developed in this study to store, manage, and provide access to student theses in digital format at Life University.
2. Thesis: refers to research project or long essays that student ready for graduate of Bachelor's degree.
3. Digital PDF Format: refers to documents that aim of being used on screen and digital devices.
4. Administrator: refers to the authorized user responsible for managing user accounts, system settings and access permissions.
5. Head of Department (HoD): refers to the academic staff member responsible for managing thesis publishing, editing, and department-related records in the system.
6. Publication Request: refers to a student's request to publish their final thesis in the repository after passing, pending review and approval by an Administrator or Head of Department (HoD).
7. Plagiarism: refers to using someone else's work, ideas, words, or research without giving proper credit or citation.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Core Concepts

2.1.1 Concept of Thesis

Thesis is a formal academic research project completed by a student in the end of a degree program. This research project gives students the opportunity to work independently on a research topic and serves as a presentation of a student's research, including the analysis and how students explore the topic, which move from general observations and discussion to facts, analysis and an overall conclusion (Tufts, 2024).

The purpose of a thesis is allow students to demonstrate the knowledge and skill they have gained during their studies, contribute to academic knowledge and by sharing research findings for future use, and help develop professional skills. The Owen Library at CED also supports the preservation and accessibility of theses for current and future users (Mentor, 2025). Through the thesis process, students develop skills in research, critical thinking, data analysis, problem solving, and academic writing which valuable for both academic and professional skills (Harvard Extension School, n.d.).

2.1.2 Digital Repository

A Digital repository (DR) is an accessible web database designed to preserve and provide access to research materials produced by an institution community. DR serve as a tool sharing academic knowledge and open access available to all users anywhere at any time (Hagen, 2009).

DR is also a collection of digital resources that allows the insertion, deletion, and retrieve of content. The main purpose of a DR is to preserve digital resources for long-term access and to provide organized retrieval of study materials without loss or damage (Dimitrios A. et al., 2004). DR sharing knowledge to the students, lectures, researchers

and the public by making research freely available online and increase the visibility of university research, preserves academic output and strengthens the institution's reputation (Ashikuzzaman, 2020).

2.1.3 Institutional Repository

An Institutional Repositories (IR) is a digital platforms developed by university to store and manage research work such as theses, dissertations, journal articles, and conference papers (Ashikuzzaman, July 2025). These systems preserve theses as electronic files, enable searching, and provide online access for students and researchers to explore the information needed for their studies (Ashikuzzaman, May 2025). IR offer significant benefits for academic institutions by providing a secure and organized digital platform for storing and preserving research output. IR also enhance the visibility and impact of scholarly work by making it accessible to a public, increasing opportunities for citations and academic recognition (Ashikuzzaman, 2025).

2.1.4 Electronic Theses and Dissertation

An Electronic Thesis and Dissertation (ETD) is a digital version of a student's thesis and dissertation that is stored and preserved in an online repository instead of printed papers as physical documents. ETD is open access and managed by the university through cooperation between the graduate school and the university library and allow research to be stored electronically and easily access by users (DUKE, n.d.).

DR, IR, and ETDs are related concepts but different in scope and focus. A DR is a general online system designed to collect, store, preserve, and provide access to various digital research resources. While an IR is a specific type of digital repository that focuses on preserving and managing the scholarly output of a specific university or institution, such as theses, dissertations, journal, and conference papers. In the other hand, ETD is the digital versions of student theses and dissertations stored within these

repositories. In short, a DR is the general system that contains various academic material, an IR is the University or Institution-based platform, and ETD is a specific type of content managed within that system.

2.2 Existing System

2.2.1 VTechWorks – Virginia Tech

VTechWorks in the IR of Virginia Tech that stores and provides online access to scholarly works, including electronic theses and dissertations. Virginia Tech was the first university to require electronic submission of theses and dissertations in 1997, supporting the preservation and online accessibility of graduate research (Virginia Tech, n.d.).

2.2.2 Apollo Institutional Repository

Apollo Institutional Repository is the institutional repository of the University of Cambridge that stores, preserves, and provides access to research outputs, including publications, theses, research data, and other academic materials (University of Cambridge, n.d.). Apollo demonstrates the benefit of using an institutional repository to preserve and provide access to academic research.

2.2.3 CamEd Open Access Repository

In Cambodia, IRs are still developing, with academic research remain fragmented across institutions. CamEd Business School developed the CamEd Open Access Repository (CamEd-OAR) to preserve, organize, and share scholarly outputs. The repository later migrated from WordPress to DSpace to improve interoperability and accessibility (Bunthorn Yem, Reviewing the Cambodian Higher Education , 2026)

2.2.4 National University of Management (NUM) Repository

The NUM provides online access to academic resources, including doctoral theses, master's theses, master's research, and bachelor's theses through Hun Sen E-Library. The resources are organized into different categories, and some thesis

documents are accessed through Google Drive, allowing users to view the documents online (Habla, 2024). The Num E-Library provides a useful local example of organizing theses by degree level and making them available online.

2.2.5 Comparison of Existing Systems.

Existing systems demonstrate different approaches to storing, organizing, preserving, and providing access to academic research and theses. However, each system has different purposes and features based on institutional requirements. The proposed DTRP focuses specifically on completed theses at Life University and provides a publication request and approval process involving students and authorized HoDs or Administrators before adding theses to the repository.

2.3 Related Studies

2.3.1 E-Thesis Repository System at the University of Tetova

Enes Bajrami (2021) developed an e-Thesis Repository System at the University of Tetova to store and provide access to university theses. The system allowed administrators to manage thesis records and user accounts while professors could upload and update thesis information. The study also considered system performance, security, and data backup. However, the system did not provide a structured workflow for student thesis submission or approval before publication. This study is relevant because it demonstrates how a digital repository can improve thesis storage and accessibility

2.3.2 Web-Based Thesis Repository System at Respati University Yogyakarta

Ajie Wibowo Soejono (2026) developed a web-based Thesis Repository System at Respati University Yogyakarta to overcome the limitations of managing physical thesis collections. Using the Waterfall model, the system provided thesis management, metadata-based searching, digital thesis access, reading history, and usage monitoring. The system achieved a System Usability Scale (SUS) score of 85, indicating excellent

usability. The study demonstrates that a digital thesis repository can improve the organization and accessibility of completed theses. It also provides useful ideas for thesis searching, digital document access, and repository management.

2.3.3 Online Thesis Repository at Mindanao State University

The Mindanao State University- General Santos City (MSU-GSC) Online Thesis Repository was developed to preserve and provide online access to published research documents. The system included searching, downloading, bookmarking, citation generation, and feedback features for students, faculty, staff, and guests. However, thesis management was primarily handled by Faculty and Staff, and the system did not include student upload permission or departmental approval before publication. This study highlights useful repository functions while revealing opportunities for more structured administrative control (Habla, 2024) .

2.3.4 Development of Thesis Repository Application in the Faculty of Science and Technology Using Vue.js Framework

Teddy Irhansyah (2023) developed a thesis repository application for the Faculty of Science and Technology at the State Islamic University of North Sumatra. The main purpose was to store student thesis data and make published theses searchable and accessible. The system was intended to help students and lecturers identify previously published thesis titles and support academic activities such as tracking thesis records and providing data. The researchers developed the system as a web application using the Vue.js framework and adopted Rapid Application Development (RAD) methodology. The system included functions such as thesis submission, thesis validation by library staff, thesis searching, downloading, bookmarking, and view student or thesis data.

2.3.5 Summary of Related Studies (Research Gap)

The reviewed studies indicate that digital thesis repositories improve thesis

storage, searching, and online accessibility. However, they generally focus on repository management after a thesis has been uploaded or published. None of the reviewed studies fully integrate a controlled thesis publication request workflow that includes student upload permission, department review, and approval before publication. There, the Digital Thesis Repository Platform in Life University addresses this gap by combining repository management with role-based access control, thesis publication requests, and approval by authorized HoDs or Administrator.

2.4 Technologies Used

The proposed Digital Thesis Repository Platform (DTRP) uses several technologies to support system development, thesis management, user authentication, file storage, database management, and web-based access. The main technologies include PHP 8.4, Laravel Framework 13, MySQL, HTML, CSS, JavaScript, Git, and GitHub.

2.4.1 Laravel

Laravel Framework is a PHP web application framework used to develop the backend of the DTRP. Laravel provides features for routing, authentication, database management, form validation, and access control. Laravel supports important DTRP functions such as thesis publication requests, thesis approval, thesis management, user management, and role-based access control.

2.4.2 PHP

PHP (Hypertext Preprocessor) is the server-side programming language used to develop the DTRP with Laravel. PHP processes user requests, handles system logic, communicates with the database, and manages operations related to thesis publication requests, thesis records, user accounts, and document files.

2.4.3 MySQL

MySQL is the relational database management system used to store and

manage DTRP data. The database stores information such as users, departments, theses, thesis files, and thesis publication requests. MySQL supports data retrieval, updating, and organization within the repository.

2.4.4 HTML

HTML (HyperText Markup Language) is used to structure the web pages of the DTRP. HTML defines elements such as forms, buttons, headings, navigation menus, and content areas displayed on the system.

2.4.5 CSS

CSS (Cascading Style Sheets) is used to control the appearance, layout, and presentation of the DTRP. CSS supports a clear and consistent interface for students, HODs, administrators, and repository users.

2.4.6 JS

JS (JavaScript) is used to provide interactive functions within the DTRP. JavaScript supports features such as form interactions, confirmation messages, dynamic content, and user interface interactions.

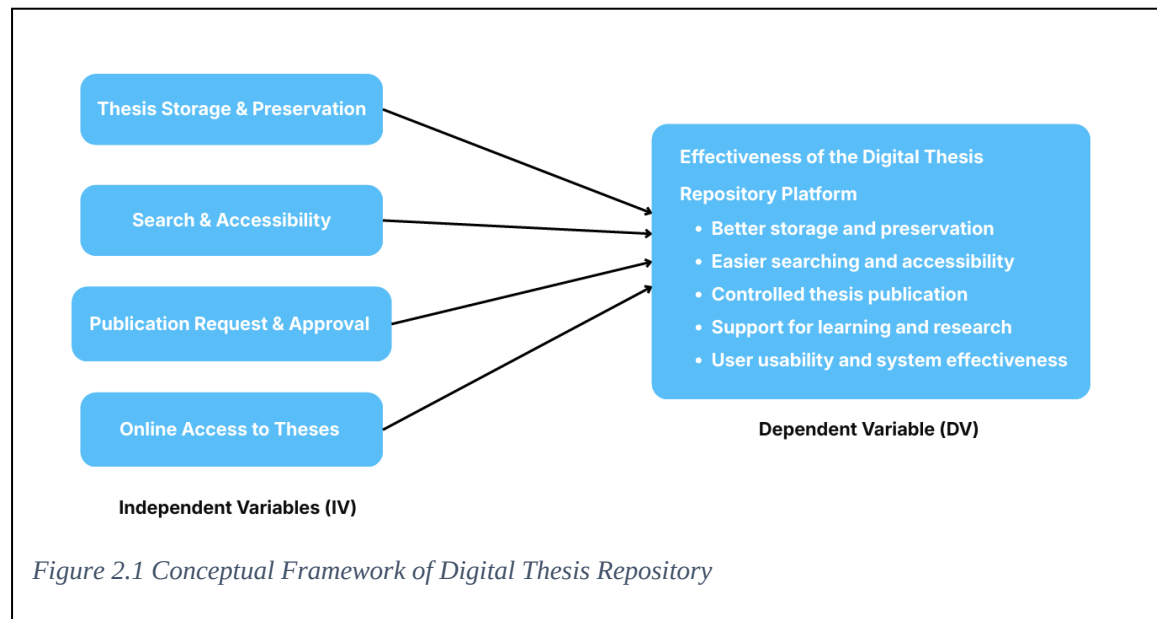
2.4.7 Git and GitHub

Git is a version control system used to track changes during DTRP development. GitHub is used to store and manage the project's source code online. Git and GitHub support branch management, version tracking, code collaboration, and project backup.

2.4.8 Mailtrap

Mailtrap is an email testing service used to test the email verification function of the system. In this project, Mailtrap is used with Laravel to receive and test verification emails during student registration. It helps ensure that verification emails are sent correctly before students can access the system.

2.6 Conceptual Framework



The conceptual framework shows the relationship between the Independent Variable (IV) and the Dependent Variables (DVs) in the proposed study. The IV is the implementation of the Digital Thesis Repository Platform (DTRP). The DTRP is expected to affect several aspects of thesis management and access at Life University.

The DVs are the expected outcomes of implementing the DTRP:

- **Storage and Preservation:** the ability to store and preserve completed theses in digital form.
- **Accessibility and Searchability:** the improvement of students and researchers to access and search previous theses.
- **Publication and Approval Process:** the ability to support students in requesting thesis publication and allow authorized HODs or administrators to approve or reject requests.
- **Support for Research and Learning:** the ability to provide approved theses as references for academic research and learning.
- **Usability and Effectiveness:** how easy, useful, and effective the

platform is for its intended users.

Therefore, the framework assumes that implementing the DTRP will improve thesis storage, accessibility, publication management, research support, and overall usability.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Methods

This study uses observation, consultation, and document and literature review to collect information and identify the requirements of the DTRP.

3.1.1 Observation

Observation is used to understand the current process of managing completed theses at LU. The researchers observe how completed theses are handled, stored and accessed. The observation helps identify existing problems and the functions needed for the proposed platform.

The researchers observe aspects such as:

1. How completed theses are currently handled and stored.
2. How students provide their completed thesis.
3. How previous theses are accessed.
4. Problems related to physical storage and searching.

3.1.2 Consultation

Consultation is used to obtain recommendations and requirements from relevant university personnel. During the presentation of the proposed research topic, the researchers discussed the proposed DTRP with the project supervisor, who is the Vice President of Life University, and received recommendations regarding the purpose, functions, and requirements of the platform. The researchers also held discussions with the Head of Department and school staff to understand how completed theses are currently managed, stored, and accessed. These recommendations are considered when defining the system requirements and designing the proposed platform.

The recommendations and discussions focused on the following areas:

1. Who should be allowed to publish a completed thesis.
2. Who should review and approve a publication request.
3. What functions should be available to students, HoDs, and administrators.
4. How completed theses are managed and stored.
5. How students submit their completed theses.

3.1.3 Document and Literature Review

Document and literature review are used to collect information from previous studies, existing repository systems, and relevant university documents. The review helps identify concepts, common features, and approaches related to digital thesis repositories.

The researcher reviews:

1. Previous studies on thesis repository systems.
2. Institutional repository studies.
3. Open-access studies.
4. Existing digital thesis repository.
5. Relevant university documents.
6. Technical documentations related to the technologies used in the DTRP.

3.2 SDLC Phases

3.2.1 Planning Phases

The planning phase is the first stage of the Software Development Life Cycle, where the researchers define the project objectives, scope and resources needed for the development of the DTRP. During this phase, the researchers identify the need for a digital platform to improve thesis storage and access at Life University. The researchers also define the project scope by identifying the target users, including Administrators, Heads of Department (HoDs), Students, and Guest users, as well as the main features to be developed. This phase provides a direction for the project and serves as a foundation

for the following requirement analysis, system design, implementation, and testing phases.

3.2.2 Requirement Analysis

The requirement analysis phase is the first stage of Software Development Life Cycle (SDLC) which focuses on identifying the system problems and determining the functional and non-functional requirements needed for the development of the system. In this phase, researchers analyze the current thesis management practices at LU and identify the needs of users who will interact with the system.

The researchers identify problems in the current thesis management process, such as difficulties in storing and searching physical theses, limited access to previous theses, risk of document damage or loss, and the need for a controlled thesis publication and approval process. The researchers then gather information, requirements, and recommendations from stakeholders to analyze the problems and determine suitable solutions.

3.2.3 System Design

The system design phase focuses on planning the structure and components of the DTRP for LU. In this stages, the researchers translates the system requirements identified in the previous phase into a clear design to guide the development process including defining the system architecture, designing the database, and planning the user interface before starting the implementation phase. The system architecture describes the interactions between users and the data flow in the system. The database design organizes thesis information, user data and system records to ensure efficient data storage and retrieval. In addition, the user interface is designed to provide a simple and user-friendly interaction for students, HoDs and Administrators and helps ensure that the platform operates efficiently, securely and reliably.

3.2.4 Implementation

In the implementation phase, the system starts to be develop based on the design

that created in the previous system design phase. The DTPR is developed using PHP programming language with the Laravel framework for backend development and MySQL is used as the database management system. During this phase, all the core functionalities of the system are implemented based on the user interface design in the previous stage including thesis publication, thesis search, user authentication, and authorization management, thesis viewing and downloading. This mean that the process transforms the designed system structure into a fully functional software system and bring it from design stage to the real-world system.

3.2.5 Testing

The testing phase ensures that after the system is developed, the system operates correctly and meets the required system specifications. Testing the system is conducted to identify and resolve the errors or system issues before deploying it to the users for actual use. Testing Functional will be performed to verify that all feature including user login, thesis publication, thesis verification by the Administrator or Hods, thesis search and thesis viewing in the system worked correctly according to the requirements. Through this process, the reliability of the system can be improved and the researchers can identify errors or system issues and fix them before deploying the system for actual use.

3.3 Project Schedule

During the planning phase, the researchers prepared a project schedule using Microsoft Project to organize the development activities, assign timeframes, and monitor project progress throughout the implementation of the Digital Thesis Repository Platform.

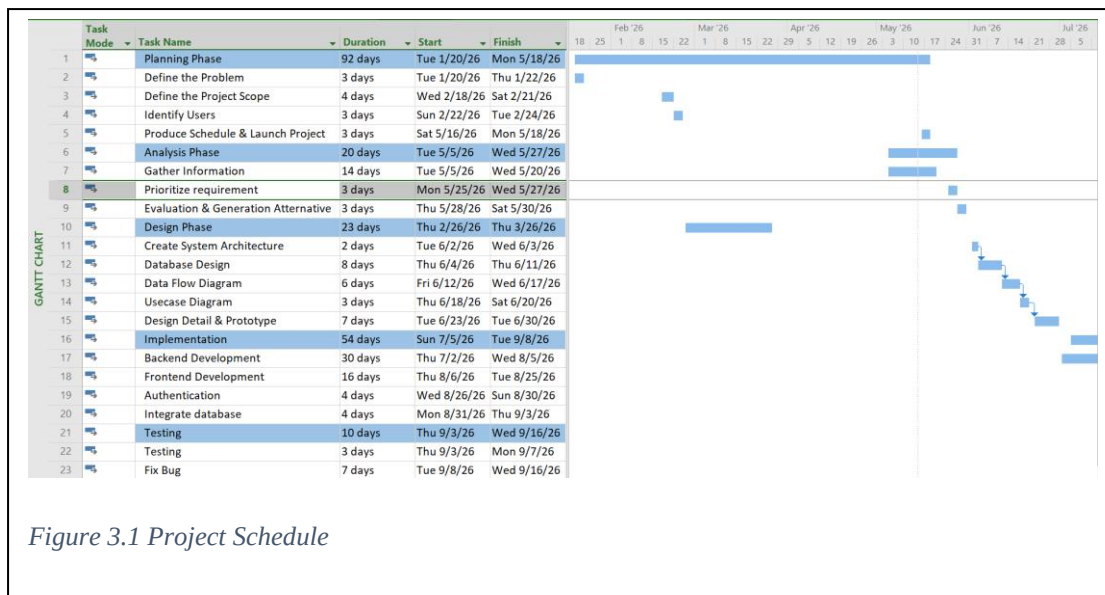


Figure 3.1 Project Schedule

3.4 System Requirement

The system requirements define the functions and qualities needed for the DTRP. The requirements are based on the identified problems, user needs, and recommendations gathered during the requirement analysis phase.

3.4.1 Functional Requirements

The functional requirements describe the main functions that the DTRP must provide. The system should:

- Allow users to log in and access features based on their roles.
- Allow the Administrator to create and manage departments.
- Allow the Administrator to manage HoD accounts and user information.
- Allow Administrator and HoDs to manage approved theses, including adding, editing, and deleting thesis information.
- Disable student upload permission by default.
- Allow Administrator and HoD to grant upload permission to fourth-year thesis team leaders (student).
- Allow students who have passed their final defense to submit thesis publication requests.

- Allow Administrator and HoDs to review, approve, or reject thesis publication requests from students.
- Allow the system to publish an approved thesis in the repository and notify the student.
- Prevent rejected thesis publication requests from being published in the repository.
- Allow students to search, view, and download approved theses.
- Allow users to search and filter theses based on information.
- Allow the system to organize theses by title, author, department, and year.

3.4.2 Non-Functional Requirements

The non-functional requirements describe the quality and performance expected from the DTRP. The system should:

- Restrict system functions according to user roles.
- Provide a user-friendly interface.
- Respond to user requests within an acceptable time.
- Store thesis information and files accurately and consistently.
- Allow the system to be updated and errors to be fixed when necessary.

3.5 System Design

3.5.1 System Architecture

The system architecture of DTRP consists of three main components including the frontend, backend, and database that interact with each other. Users such as students, guest and researchers, administrators and Hods interact with the system through the frontend interface developed using HTML and CSS and then frontend communicates with the backend built using the Laravel framework, which processes user request, manages system logic, and controls system functions. All thesis records, user accounts

and system data will be stored and managed in the MySQL database.

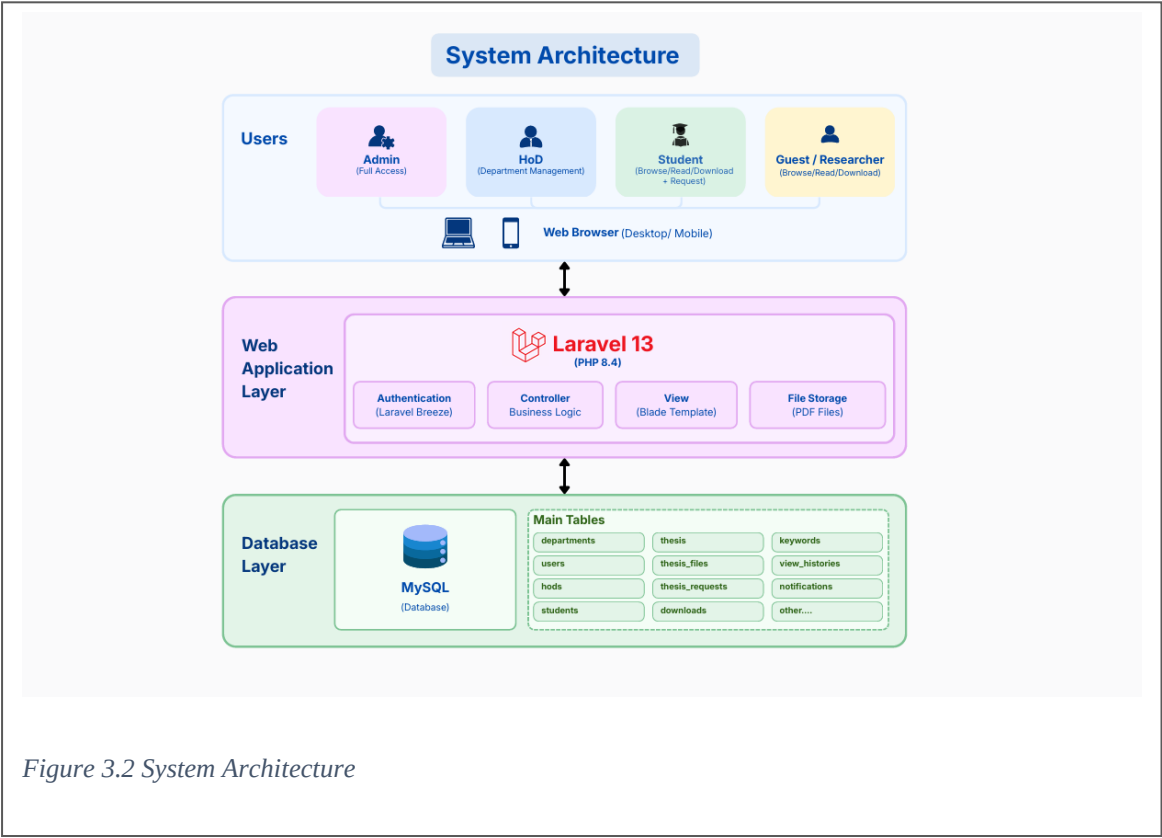


Figure 3.2 System Architecture

3.5.2 Context Diagram

The context diagram indicates the overall interaction the overall interaction between external users and the DTRP. In this system, there are four main external entities interact with the platform including administrator, Hods, student with upload permission and general students or researchers. In this interactions, students with permission can search, view, download theses and send upload request through the system since the general students or researchers are allowed to search, view and download theses for reference purposes. HoDs are responsible for reviewing and approving or rejecting thesis requests and can upload thesis documents from their departments for public reading too. Meanwhile, administrators manage Hod accounts and monitor system reports. The context diagram provide a view of how external users exchange information with the DTRP system. The following figure was prepared by the researchers to indicate how external entities interact with the platform:

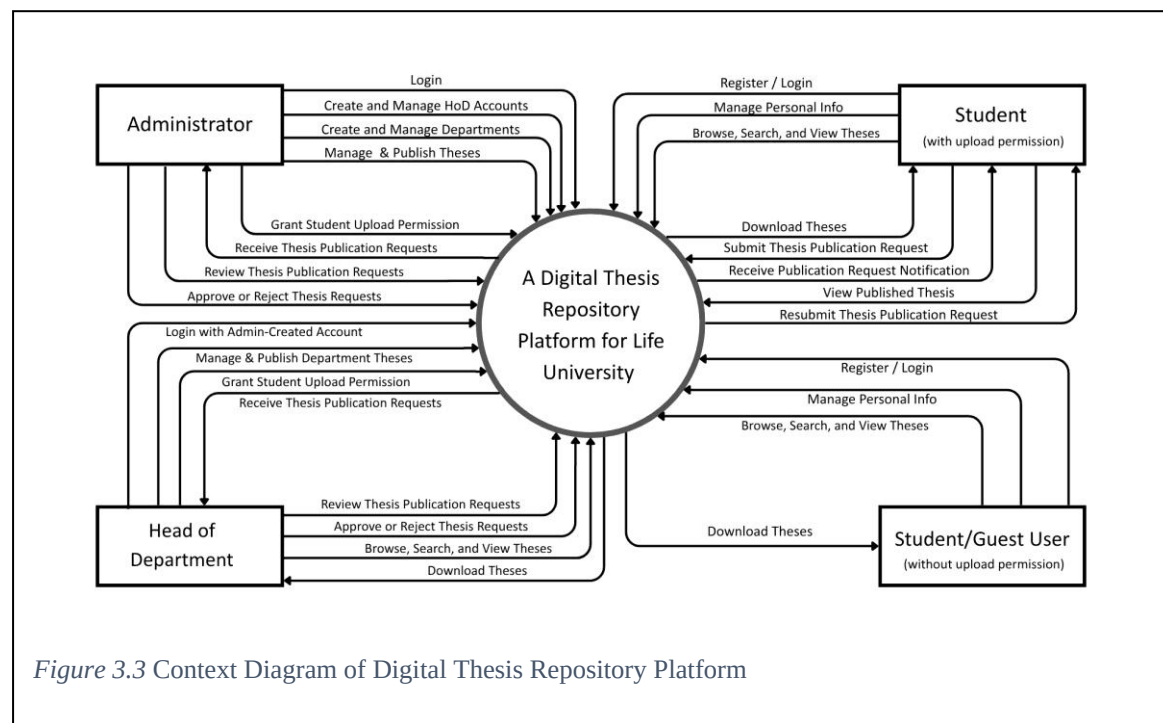


Figure 3.3 Context Diagram of Digital Thesis Repository Platform

3.5.2.1 Data Flow Diagram of Digital Thesis Repository Platform

(Level 0)

As discussed earlier, the Administrator is responsible for managing HoD accounts and department information within the system. Both the Administrator and HoD can grant upload permission, review students' thesis publication requests, and publish theses to the repository. However, the HoD is only allowed to review requests and publish theses within their respective department. Students with upload permission can submit a thesis publication request for their completed thesis and wait for the approval decision. Students without upload permission and guest users can search, view, and download published theses. The following Data Flow Diagram illustrates how the Administrator, Head of Department (HoD), students, and guest users interact with the system through eight main processes.

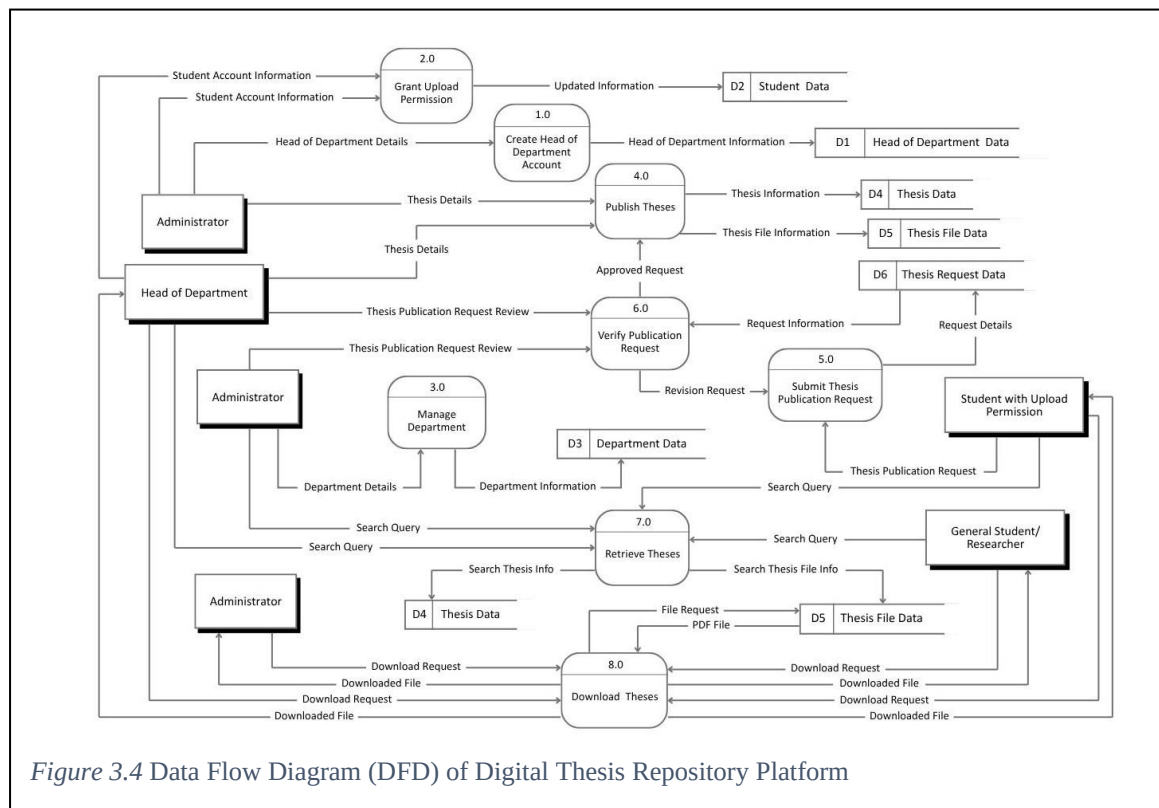


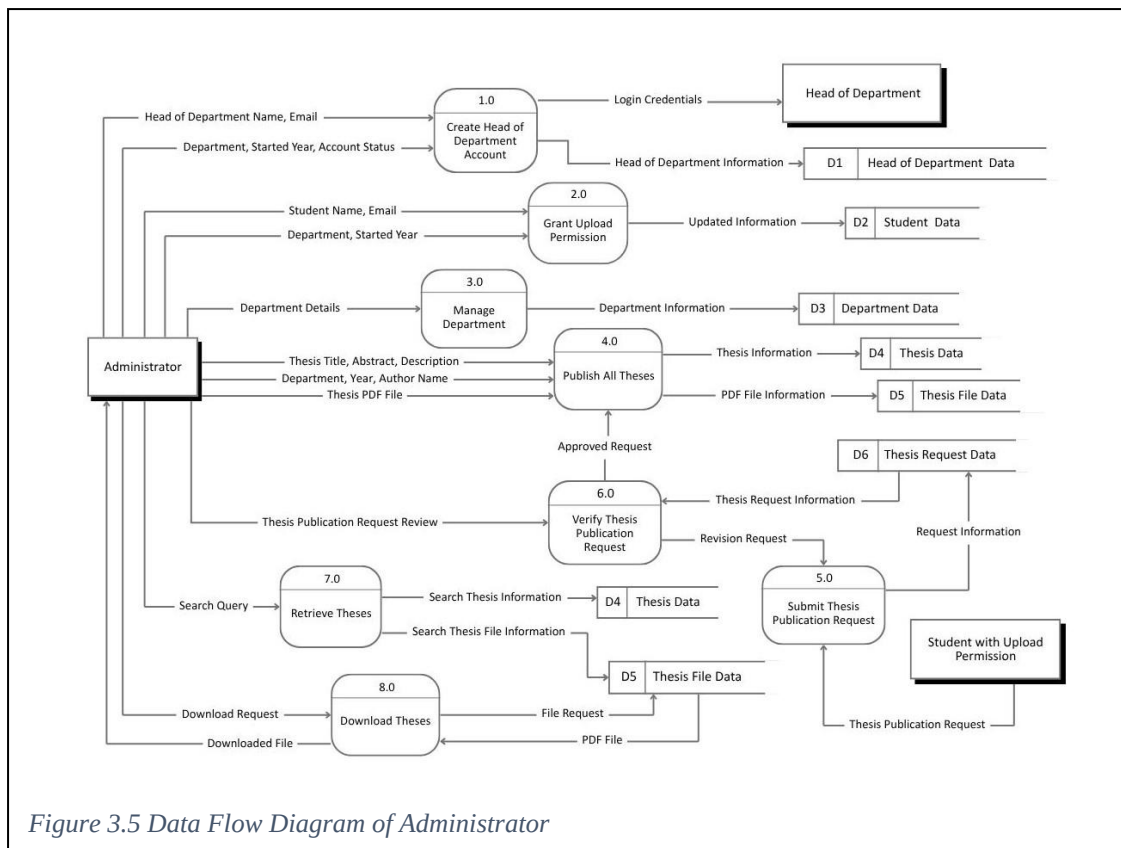
Figure 3.4 Data Flow Diagram (DFD) of Digital Thesis Repository Platform

3.5.2.2 Data Flow Diagram of Administrator

As previously described, the Administrator creates HoD accounts by entering information such as full name, email, department, year, and account status. The HoD account information is then stored in the HoD account data store, and the Administrator provides the account email and password to the respective HoD.

The Administrator manages and publishes theses from all departments by entering information such as title, abstract, description, department, year, author name, and PDF file. The thesis information is stored in the Thesis Data Store, while the PDF file is stored in the Thesis File Data Store.

Another responsibility of the Administrator is to grant upload permission to students after receiving their account information. Once permission is granted, students can submit thesis publication requests. The Administrator reviews and verifies thesis publication requests submitted by students, and either approves or rejects them. If a request is rejected, the Administrator provides comments for the student to revise and resubmit the request. If approved, the thesis is published in the repository. The Administrator can also search, view, and download published theses. The following diagram illustrates the detailed data flows and processes performed by the Administrator.

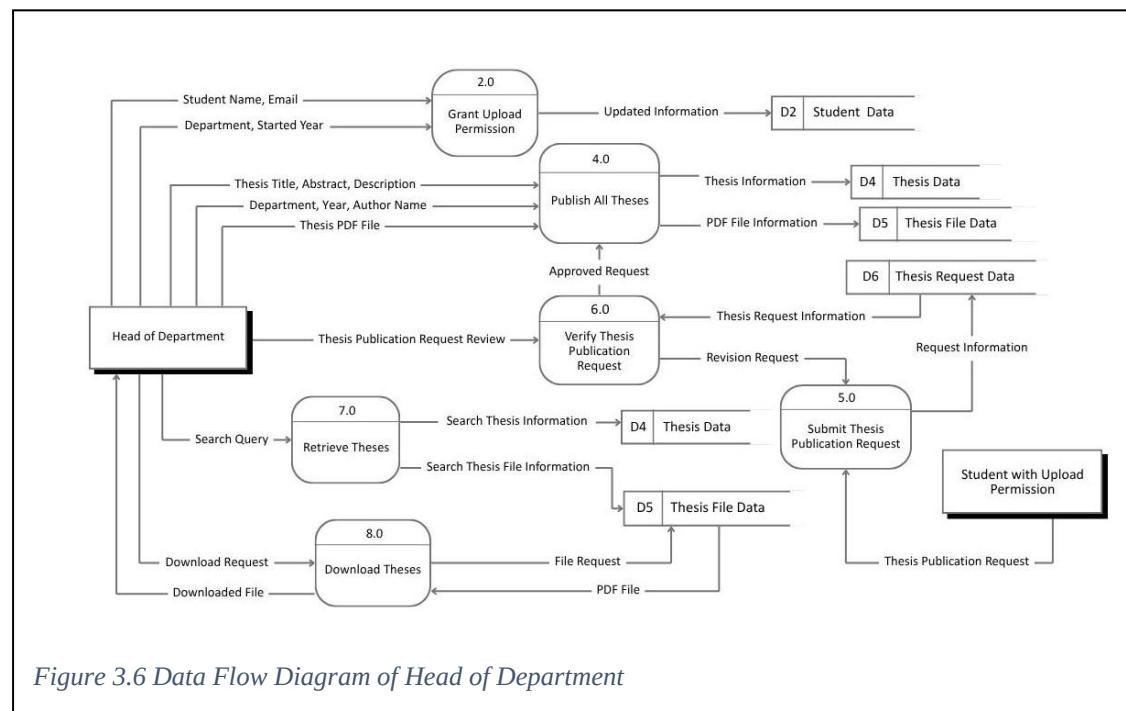


3.5.2.3 Data Flow Diagram of Head of Department

The Head of Department (HoD) is responsible for managing and reviewing theses with their respective department. Similar to the Administrator, the HoD can manage and publish theses, as well as review thesis publication requests submitted by students. However, the HoD has more limited access to system functionalities than the Administrator. The HoD does not have permission to create HoD accounts or manage HoD account information. In addition, the HoD can manage and publish theses and review thesis publication requests only within their own department.

For theses belonging to other departments, the HoD has read-only access. They can search for, view, and download published theses from other departments but cannot add, update, delete, or otherwise manage them. Therefore, the HoD has full management access to theses within their own department and read-only access to theses from other departments. The diagram below presents the data flows and processes

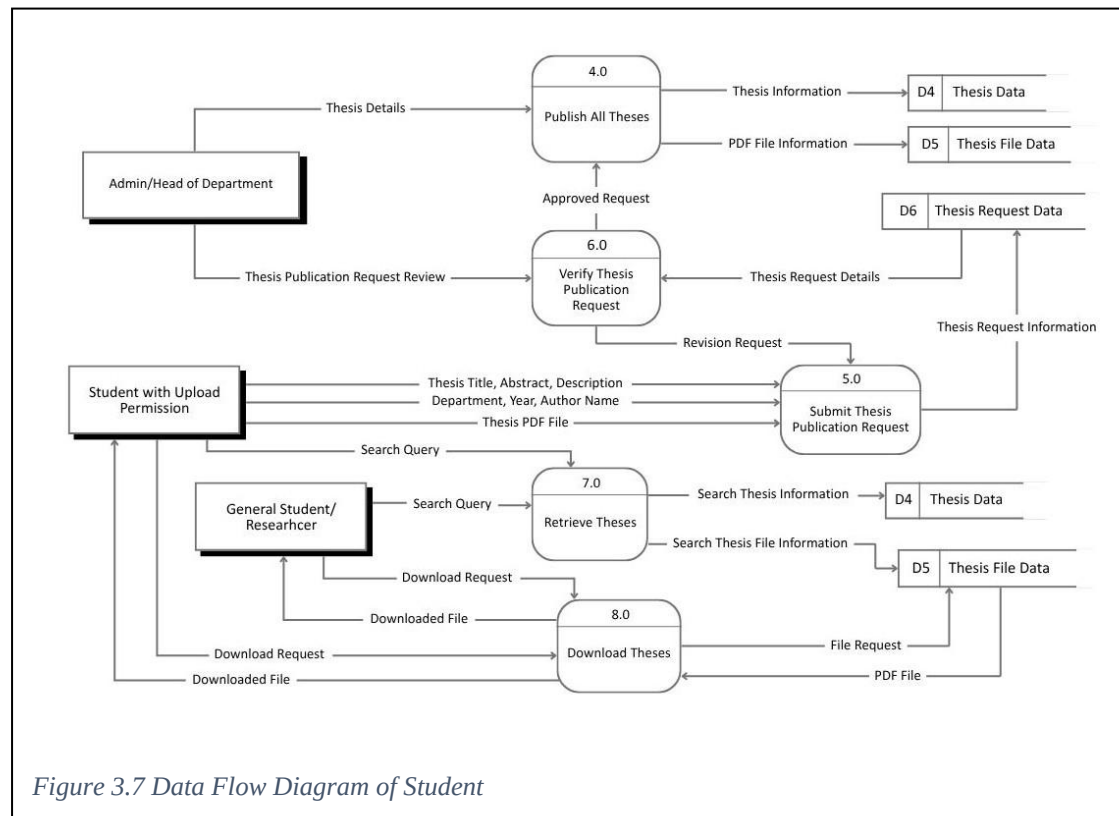
associated with the HoD's activities and illustrates the department-based access restrictions within the system.



3.5.2.4 Data Flow Diagram of Student

The Student Data Flow Diagram consists of two types of students, Students with Upload Permission and Students without Upload Permission. A Student with Upload Permission is typically the team leader of a thesis group who provides their account information directly to the Administrator or Head of Department. The Administrator or HoD then grants upload permission to the student through the system. Once permission is granted, the student can submit a thesis publication request by providing thesis information, including the title, abstract, description, year, department, author names, and the thesis PDF file. The submitted request is then reviewed by the Administrator or HoD, who either approves or rejects it. If the request is approved, the thesis is automatically published in the repository. If the request is rejected, the Administrator or HoD provides comments for the student to revise and resubmit the request. A Student without Upload Permission cannot submit thesis publication requests.

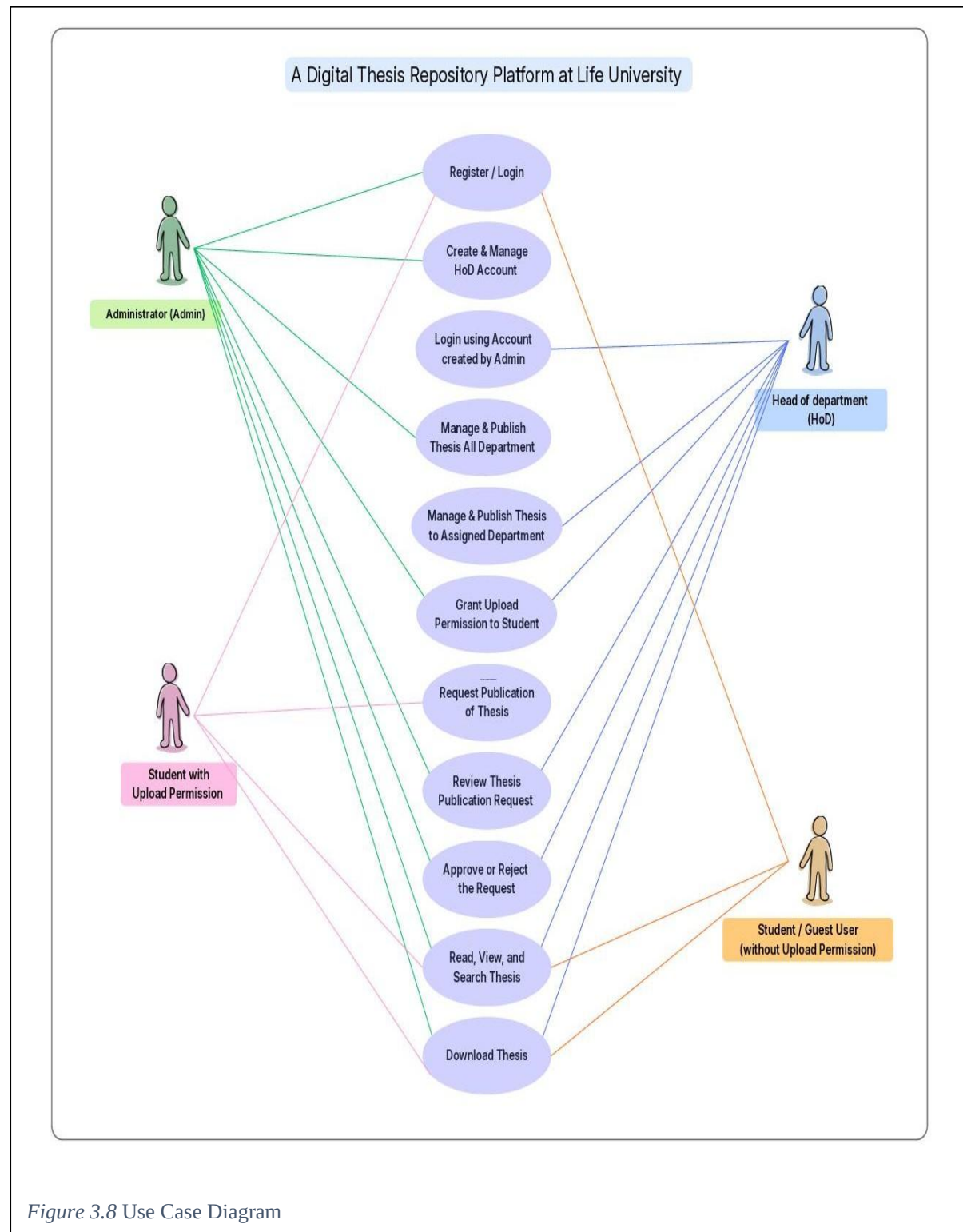
However, both types of students, as well as guest users, can search, view, read, and download published theses from the repository. The diagram below presents the data flows and processes associated with student activities and their respective access permissions.



3.5.3 Use Case Diagram

Now, let's move on to use case diagram. The use case diagram illustrates the functional interactions between users and the DTRP and show how different users perform specific actions within the system. The main actors involved in the system include administrator, Hod, and student with permission, as well as general student or guest users. As mentioned earlier, the administrator is responsible for creating Hod accounts and manage accounts to ensure security in the system. Hods review and approve student upload requests and can upload thesis documents to the repository. Students with permission can search, read, download and send upload requests to their respective HoDs within their department since the general students or guests are allowed to search, view

and download theses. The use case diagram provides a clear overview of the main functions available in the system and how each actor interacts or communicates with the platform, as shown below:



3.5.4 System Flowchart

The system flowchart begins with the administrator's process. As discussed earlier, the administrator is responsible for creating accounts for Hods and managing them to ensure proper system access. Moreover, when one of the Hod leaves the university, the administrator terminates the Hod's account to prevent the user from logging into the system again after leave. The account is not removed or deleted by administrator because it could affect the integrity of the database. Instead of deleting the accounts, the researchers decided to allow administrator to terminate the login, ensures that former Hods no longer have access to the system while maintaining all the historical records and data.

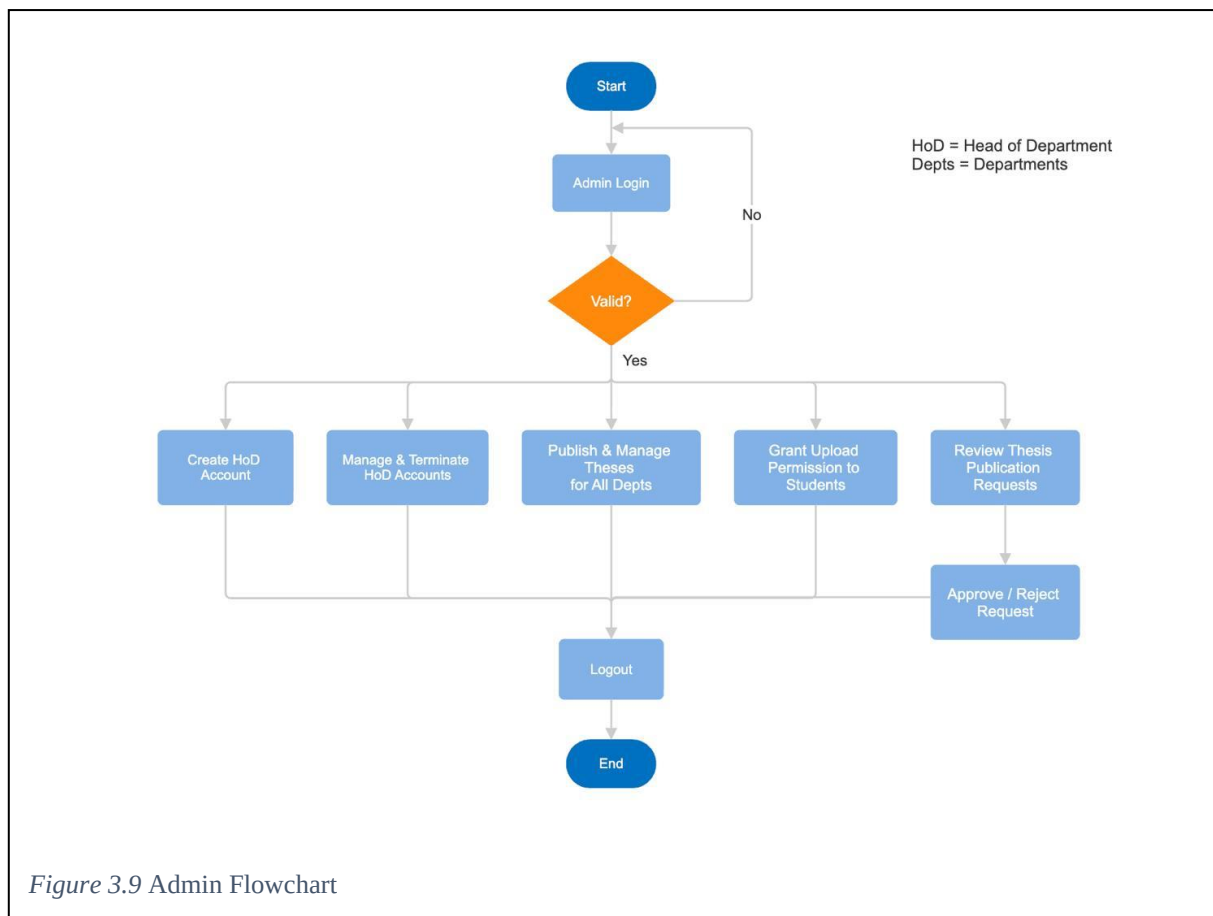
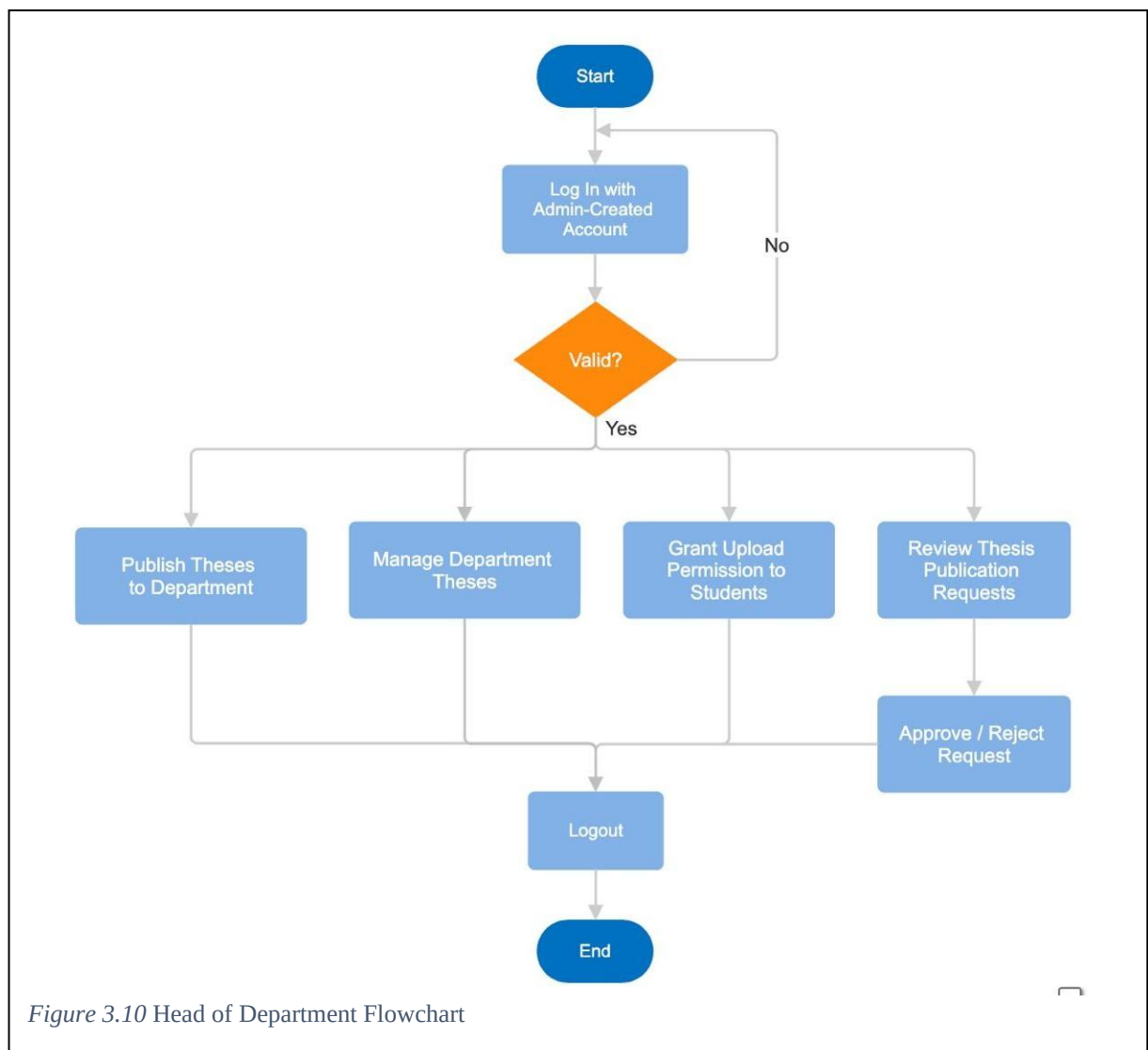


Figure 3.9 Admin Flowchart

After the admin flowchart, the process continues with the Hod who is responsible for managing student thesis uploads within their department. When a student submits a request to upload their thesis, Hod reviews the request to ensure that the document is thesis research work and it is correct and follows the required thesis format at LU. Once verified, the Hod approves the upload then thesis will be allow to be published in the DTRP. In this process, allowing Hod to review theses ensures that all documents are properly checked before publication to students, guests, and other user and maintaining the quality of the repository.



Now, let's continue to the student flowchart. Students can interact with the DTRP by searching, viewing and downloading thesis from the repository for reference purposes. Moreover, when a student needs to upload their thesis after passing the final defense, the student have to send upload request documents to their respective Hod through the system first. Then, once the request is approved by the Hod, the student's thesis is publish in the repository. This process ensures that students have controlled access to upload their thesis while maintaining the security of the repository.

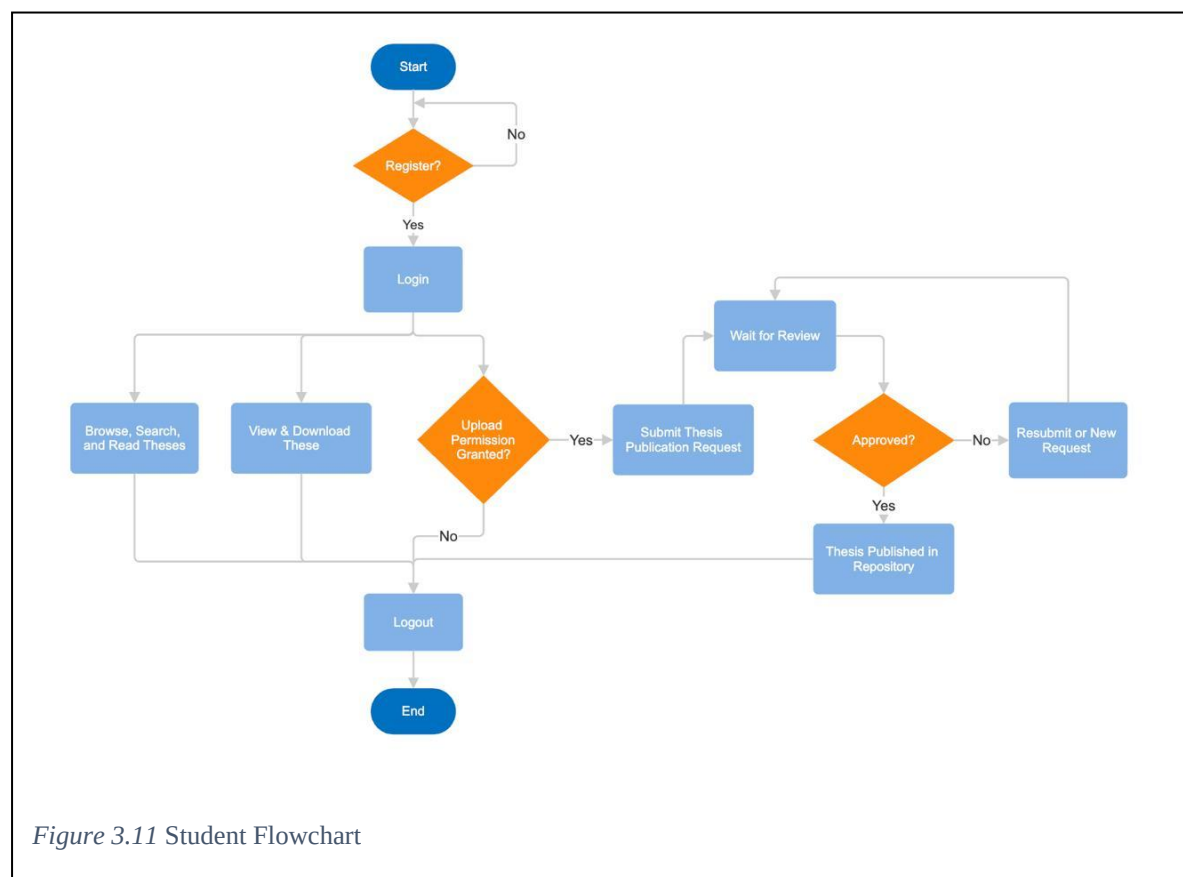


Figure 3.11 Student Flowchart

3.5.5 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) represents the database structure of the DTRP for LU, and it illustrates how data are organized and how they are related to each other in the system. The database of this system consists of thirteen main tables, including User, Department, Student, Hod, Theses, Thesis_file, Keyword, Thesis_keyword, Saved_thesis, Viewed_History, Thesis_Request, Download, and Notifications. All of these tables store important information such as user accounts, departmental data, thesis records, thesis files, and download history, which are necessary for the system. The relationship between these tables allows the system to manage thesis documents and track user interactions with the system, such as viewing and downloading theses. The following figure illustrates the overall database organization and the table relationships between the tables:

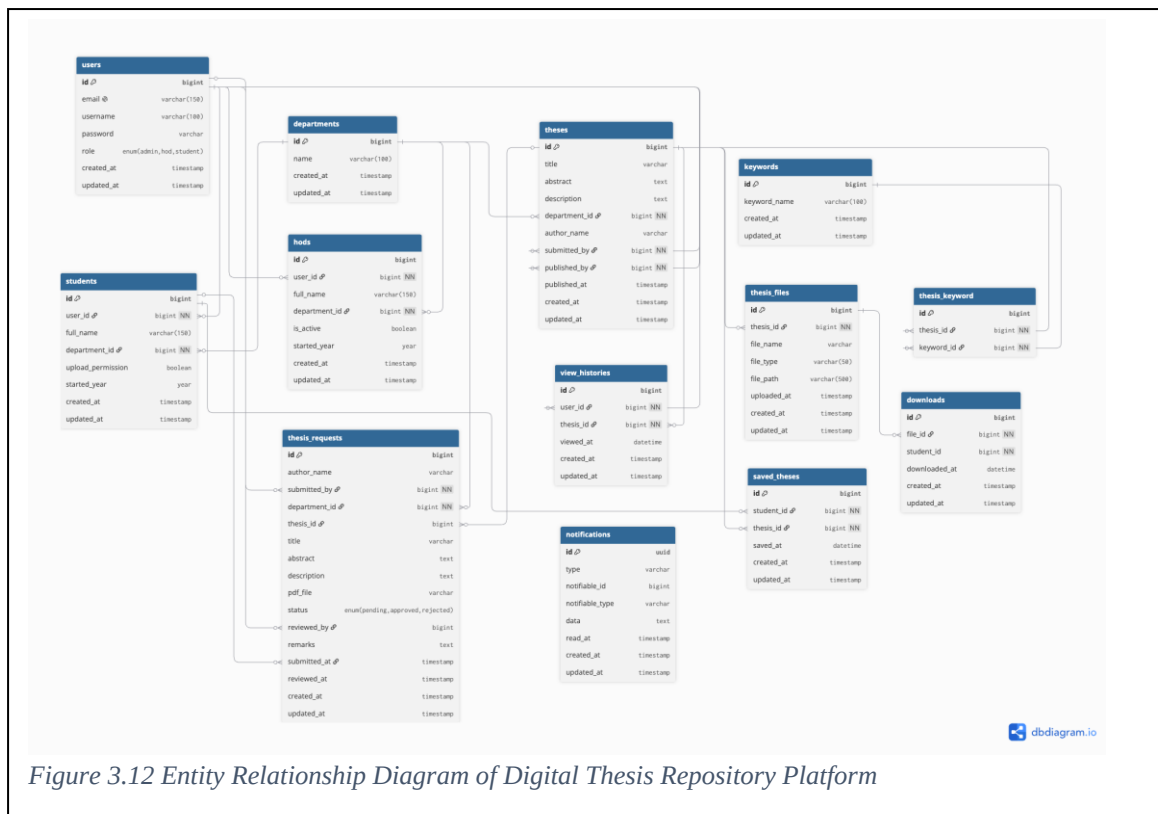


Figure 3.12 Entity Relationship Diagram of Digital Thesis Repository Platform

3.5.6 Database Design

The Database Design is developed to provide detailed descriptions of the entities, attributes, and relationships presented in the ERD in the previous section. It defines the structure of each table in the database, including attribute names, data types, and descriptions, to help developers and system users understand how data is organized. The details are shown in the following tables:

Table Name	Field Name	Datatype	Require	Key	Default	Reference	Description
Department	Id	int	Y	PK	Auto Increment		Unique ID of Department
	Name	varchar	Y	-	Not NULL		Name of Department
	created_at	timestamp	Y	-	current_timestamp		Record creation date
User	id	int	Y	PK	Auto Increment		Unique ID of user
	email	varchar	Y		Unique		User Email
	username	varchar	Y		Not Null		Login Username
	password	varchar	N		Not Null		User Password (encrypted)
	role	enum	Y		Student		Role of User (admin, hod, student)
	created_at	timestamp	Y	-	current_timestamp		Record creation date
Student	id	int	Y	PK	Auto Increment		Unique ID of Student
	user_id	int	Y	FK	Not Null	<u>users.id</u>	Linked User Account
	full_name	varchar	Y		Not Null		Student email
	department_id	int	Y	FK	Not Null	<u>departments.id</u>	Linked to Department Table
	upload_permission	boolean	Y		False		Permission to Request Thesis Publication
	started_year	year	Y		-		Academic Year of Student
	created_at	timestamp	Y		current_timestamp		Record creation date
Head of Department	id	int	Y	PK	Auto Increment		Unique ID of Head of Department
	user_id	int	Y	FK	Not Null	<u>users.id</u>	Linked User Account
	full_name	varchar	Y		Not Null		Student Email
	department_id	int	Y		Not Null	<u>departments.id</u>	Linked to Department Table
	Is_active	boolean	Y		True		Manage Account Access
	started_year	year	Y		Null		Stared Year of Head of Department
	created_at	timestamp	Y		current_timestamp		Record creation date
Thesis	id	int	Y	PK	Auto Increment		Thesis ID
	title	varchar	Y		Not Null		Thesis Title
	abstract	text	Y		Not Null		Thesis Abstract
	description	text	N		Not Null		Description of Thesis
	department_id	int	Y	FK	Not Null	departments.id	Linked to Department Thesis
	author_name	varchar	Y		Not Null		Author(s) Name of Thesis
	submitted_by	int	Y	FK	Not Null	<u>users.id</u>	User who submitted the thesis
	published_by	int	Y	FK	Null	<u>users.id</u>	User who published the thesis
	published_at	timestamp	Y		current_timestamp		Thesis Publication Date
	created_at	timestamp	Y		current_timestamp		Record creation date

Thesis File	id	int	Y	PK	Auto Increment		Unique ID of Thesis File
	thesis_id	int	Y	FK	Not Null	<u>theses.id</u>	Linked to Thesis Table
	file_name	varchar	Y		Not Null		Thesis File Name
	file_type	varchar	Y		Not Null		File Format
	file_path	varchar	Y		Not Null		Storage Path
	uploaded_at	timestamp	Y		current_timestamp		Record Upload Date
	created_at	timestamp	Y		current_timestamp		Record creation Date
Download	id	int	Y	PK	Auto Increment		Download ID
	file_id	int	Y	FK	Not Null	<u>thesis_files.id</u>	Linked to Thesis File Table
	student_id	int	Y	FK	Not Null	<u>students.id</u>	Linked to Student Table
	downloaded_at	DateTime	Y		Not Null		Download Date
	created_at	timestamp	Y		current_timestamp		Record creation date
Keyword	id	int	Y	PK	Auto Increment		Keyword Id
	keyword_name	varchar	Y		Not Null		Keyword Name
	created_at	timestamp	Y		current_timestamp		Record creation date
Thesis Keyword	id	Int	Y		Auto Increment		Thesis Keyword Id
	thesis_id	int	Y		Not Null	<u>theses.id</u>	Linked to Thesis Table
	keyword_id	int	Y		Not Null	<u>keywords.id</u>	Linked to Keyword Table
	thesis_id	Int	Y		Not Null		Unique ID of thesis and keyword
Saved Thesis	id	int	Y	PK	Auto Increment		Saved Thesis ID
	student_id	int	Y	FK	Not Null	<u>student.id</u>	Linked to Student Table
	thesis_id	int	Y	FK	Not Null	<u>theses.id</u>	Linked to Thesis Table
	saved_at	DateTime	Y		Not Null		Record Save Date
View History	id	int	Y	PK	Auto Increment		View History Id
	user_id	int	Y	FK	Not Null	<u>users.id</u>	Linked to User Table
	thesis_id	int	Y	FK	Not Null	<u>thesis.id</u>	Linked to Thesis Table
	viewed_at	DateTime	Y		Not Null		Record View DateTime
Thesis Request	id	int	Y	PK	Auto Increment		Thesis Request Id
	author_name	varchar	Y		Not Null		Author Name of Thesis Request
	submitted_by	int	Y	FK	Not Null	<u>users.id</u>	Student who submitted Request
	department_id	int	Y	FK	Not Null	<u>departments.id</u>	Department of Thesis Request
	thesis_id	int	N	FK	Null	<u>theses.id</u>	Linked to Thesis Table after Publish
	title	varchar	Y		Not Null		Thesis Request Title
	abstracts	text	Y		Not Null		Thesis Request Abstract
	description	text	Y		Not Null		Thesis Request Description
	pdf_file	varchar	Y		Null		Thesis Request PDF File
	status	enum	Y		Pending		Thesis Request Status ['Pending', 'Approved', 'Rejected']
	reviewed_by	int	N	FK	Null	<u>users.id</u>	User who Reviewed the Request
	remarks	varchar	N		Null		Feedback after Reject Request
	submitted_by	timestamp	Y		current_timestamp		Record of Submitted Date
	reviewed_at	timestamp	N		Null		Record of Reviewed Date
	created_at	timestamp	Y		current_timestamp		Record creation date
Notification	id	int	Y	PK	Auto Increment		Unique notification ID
	type	varchar	Y		Not Null		Type of notification

	notifiable	morphs	Y		Not Null		Notifiable entity ID and type receiving the notification
	date	text	Y		Not Null		Notification data
	read_at	timestamp	N		Null		Record of Read Date
	created_at	timestamp	Y		current_timestamp		Record creation Date

Table 3.1 Database Design of Digital Thesis Repository Platform

3.6 Tools and Technologies

As discussed in Section 2.5, the DTRP adopts several tools and technologies to support the system design. Laravel with PHP is selected to implement the backend architecture, including authentication, role-based access control, thesis management, and publication request processing. MySQL is chosen to store and organize thesis records, user accounts, departments, and other system data through a relational database structure. HTML, CSS, and JavaScript are used in the proposed interface to provide a responsive and user-friendly web experience. These technologies are selected because they support the functional requirements of the DTRP and provide a reliable foundation for developing a maintainable and scalable web-based repository system.

3.7 Summary

This chapter describes the research methodology and the system design used in the study. It explains the system architecture, database design, data flow diagram, and the Entity Relationship Diagram (ERD) along with the data dictionary that defines the detailed database structure. In addition, the chapter mentioned the tools and technologies used in developing the Digital Repository Platform (DTRP), including MySQL, PHP with the Laravel framework, and frontend technologies such as HTML, CSS, and JavaScript to build reliable web-based system for managing thesis documents at LU.

CHAPTER 4

SYSTEM IMPLEMENTATION AND TESTING

4.1 Development Environment

The development of the DTRP was carried out using a combination of hardware devices and software tools to support system development, testing, and deployment. The project was developed by three team members using both Windows and macOS. Laragon was used as the local development environment on Windows, while Laravel Herd was used on macOS to ensure compatibility across different development platforms.

4.1.1 Hardware Requirements

The DTRP was developed using multiple devices with sufficient hardware specifications to support web application development, database management, and system testing. The development team used ASUS laptops and MacBook running Window and macOS operating systems. These devices provided processing power, memory, and storage for running Laravel, MySQL, local development environments, and testing tools throughout the development process.

4.1.2 Software Requirements

The Development of the DTRP required several software tools to support coding, database management, version control, and system testing. Visual Studio Code was used as the primary course code editor, while Laravel 13 and PHP 8.4 were used for web application development. MySQL served as the database management system, and Laragon and Laravel Herd were used as local development environments for Window and macOS. Git and GitHub supported version control and collaboration development, while Google Chrome was used to test and verify the functionality of the implemented system.

4.2 System Implementation

4.2.1 Welcome Page

The Welcome Page is the first page of the Digital Thesis Repository Platform. It introduces the purpose of the system and provides access to the Login and Register pages for students and guest users. Public visitors can also browse published theses directly from this page before accessing other system features.

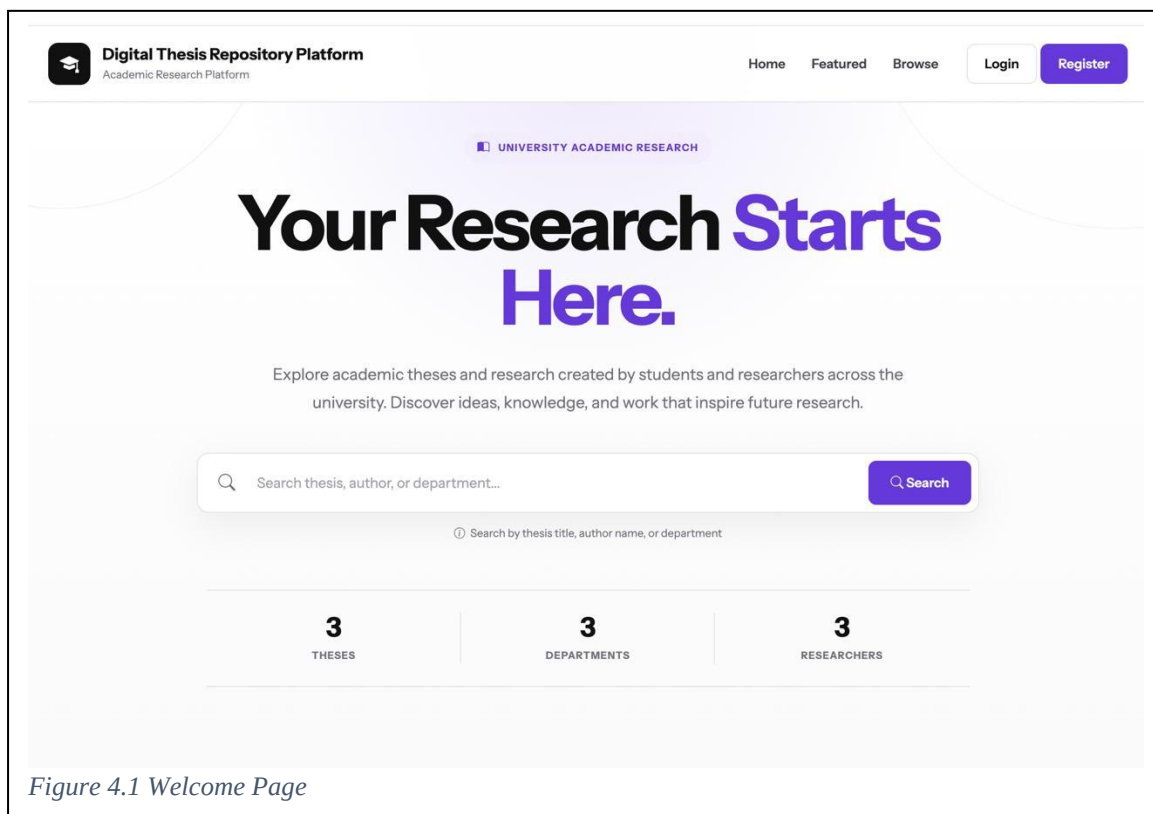


Figure 4.1 Welcome Page

4.2.2 Authentication

4.2.2.1 User Registration

The User Registration feature allows new users to create an account by providing information such as full name, username, email, and password. When a user enters a Life University email address (e.g., abc@lifeun.edu.kh), the registration form displays additional fields for Department and Start Year. These fields are not displayed for guest users who register with external email addresses (e.g., abc@gmail.com). After registration, users must verify their email before accessing the system. Based on the

registered email domain, the system identifies the user as either a Life University student or a guest user.

The User Registration feature is available only for students and guest users. The system creates the Administrator account, while HoD accounts are created by the Administrator. Therefore, Administrators and HoDs do not register through the registration page and can only log in using their assigned accounts

Create Account

Please fill in your information to register.

Full Name

Username

Email Address

Password

Confirm Password

[Create Account](#)

Already have an account? [Login](#)

Figure 4.2 Guest Register Page

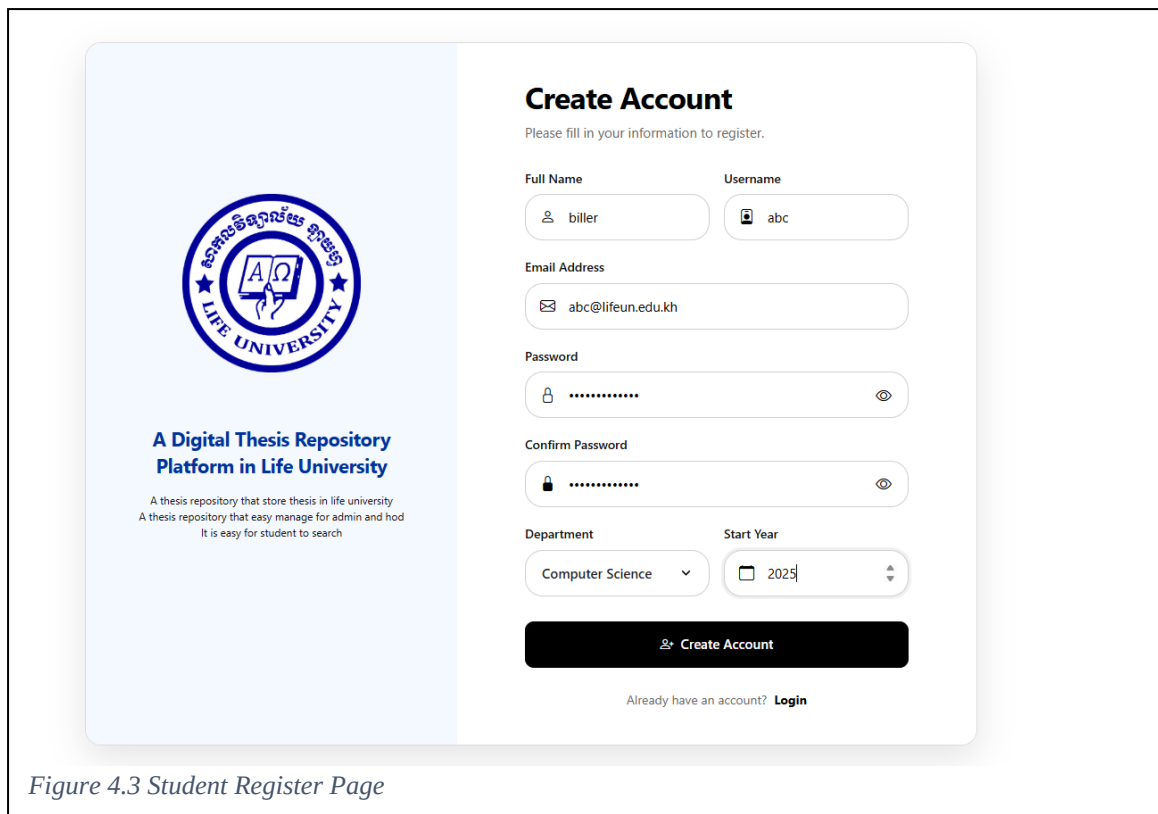


Figure 4.3 Student Register Page

4.2.2.2 Email Verification

After completing the registration process, the system sends a verification email to the user's registered email address. Users must click the verification link before they can log in and access the system. This feature is applied to both Life University student accounts and guest user accounts to ensure that only verified email addresses can use the system. During the development and testing process, the researchers used Mailtrap as the email service to send and test verification emails in a development environment. This allowed the email verification process to be tested without using a production email service.

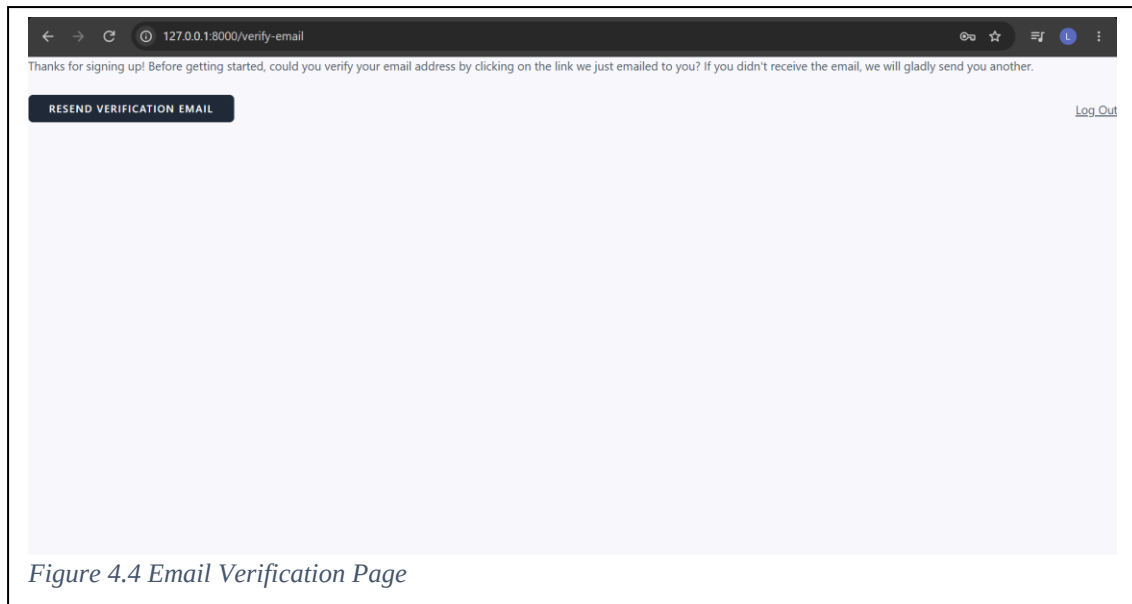


Figure 4.4 Email Verification Page

4.2.2.3 User Login

The User Login feature allows registered and verified users to access the system using their email and password. The system verifies the login credentials and redirects users to the appropriate dashboard based on their assigned role, such as Administrator, HoD, Student, or Guest User.

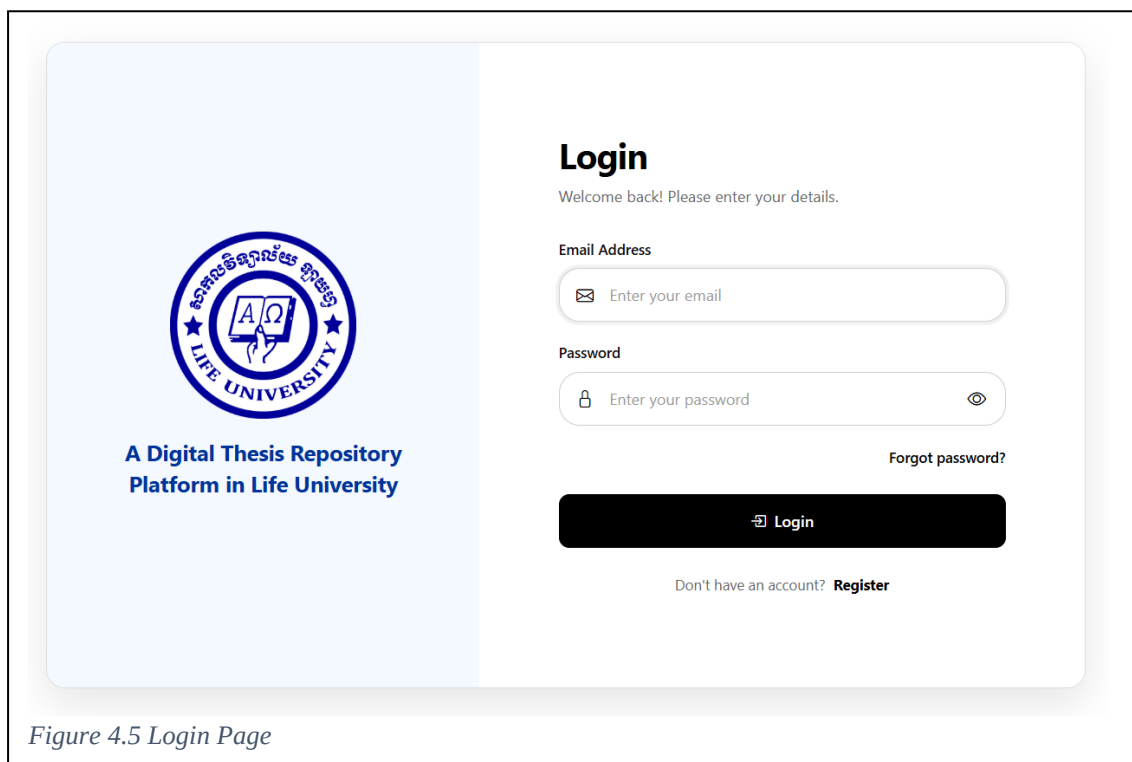


Figure 4.5 Login Page

4.2.2.4 Role-Based Access

The User Login feature allows registered and verified users to access the system using their email and password. The system verifies the login credentials and redirects users to the appropriate dashboard based on their assigned role, such as Administrator, HoD, Student, or Guest User.

4.2.3 Administrator Module

4.2.3.1 Administrator Dashboard

The Administrator Dashboard provides an overview of the system and its main activities. It displays information such as the total number of departments, students, theses, and pending requests for thesis publication. The dashboard also provides quick access to the Administrator's main functions.

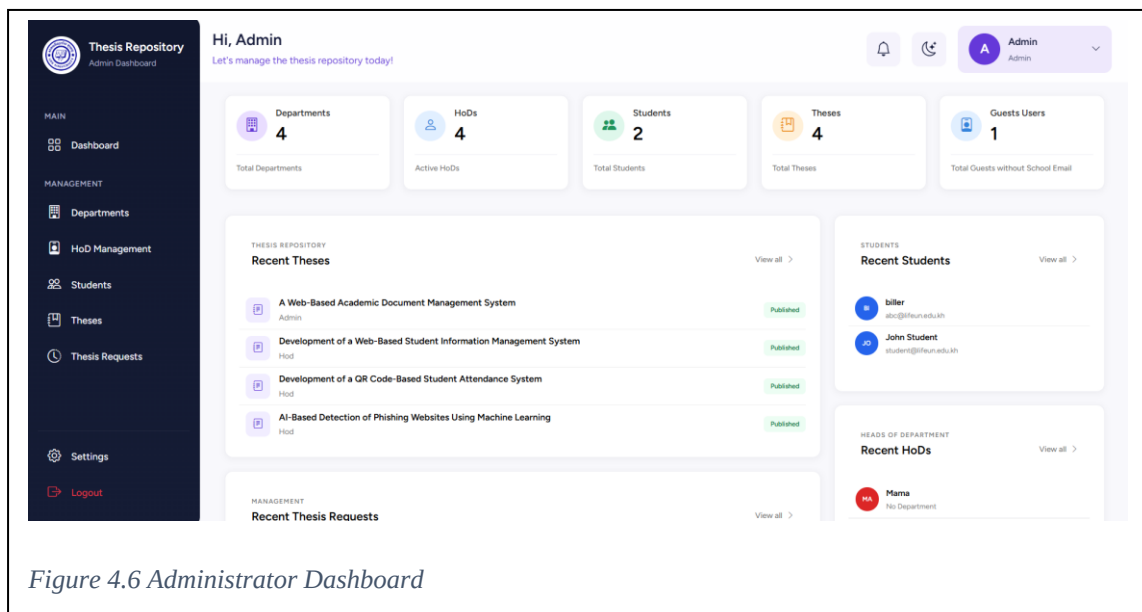


Figure 4.6 Administrator Dashboard

4.2.3.2 Department Management

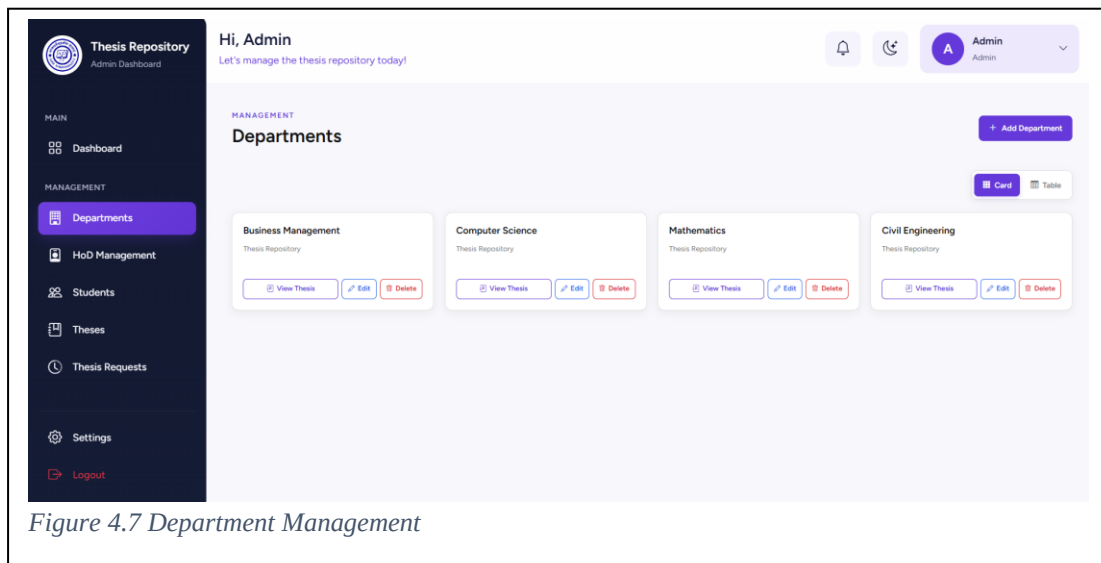


Figure 4.7 Department Management

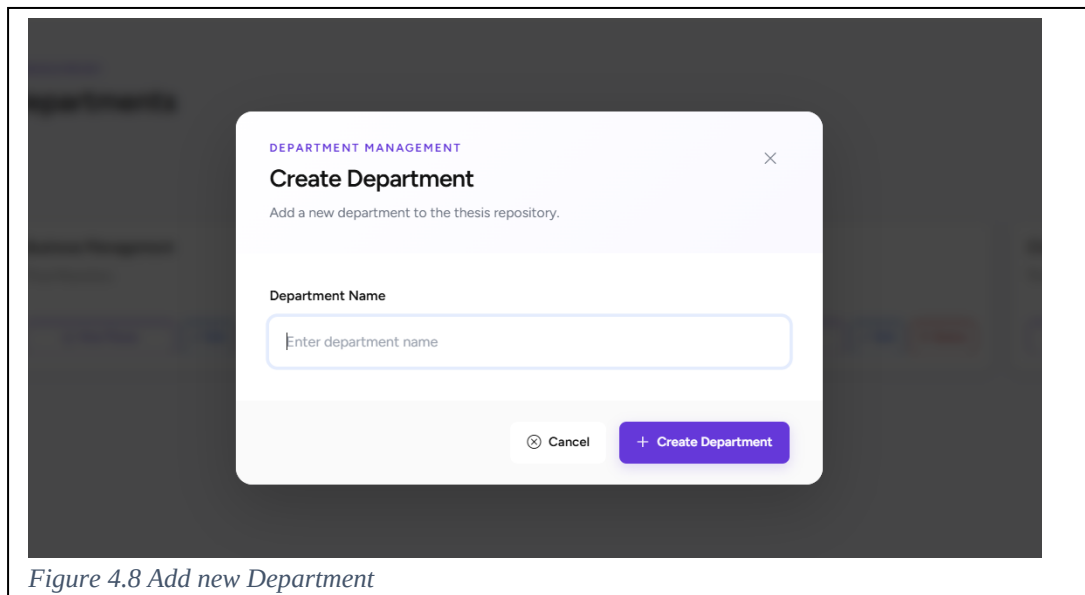


Figure 4.8 Add new Department

4.2.3.3 HoD Account Management

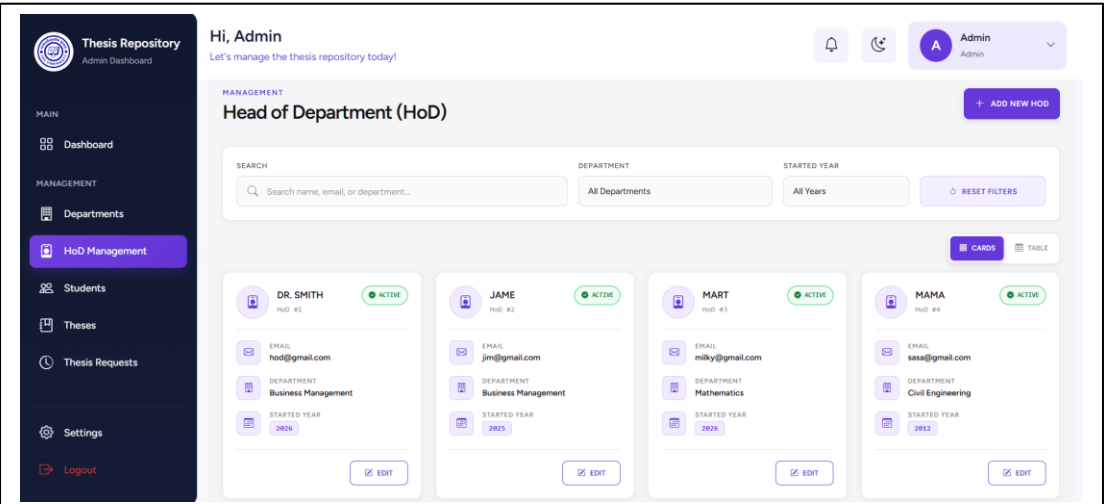


Figure 4.9 HoD Management in Card View

The screenshot shows the 'Add Head of Department' form in the 'Thesis Repository Admin Dashboard'. The form is titled 'Add Head of Department' and includes the instruction 'Create a new Head of Department account.' The form fields are:

- FULL NAME: Enter full name
- EMAIL: Enter email address
- USERNAME: Enter username
- PASSWORD: Enter password
- DEPARTMENT: Select Department (dropdown menu)
- STARTED YEAR: 2025
- STATUS: Active

At the bottom right of the form are two buttons: 'CANCEL' and 'CREATE HOD'.

Figure 4.11 HoD Account Creation Form

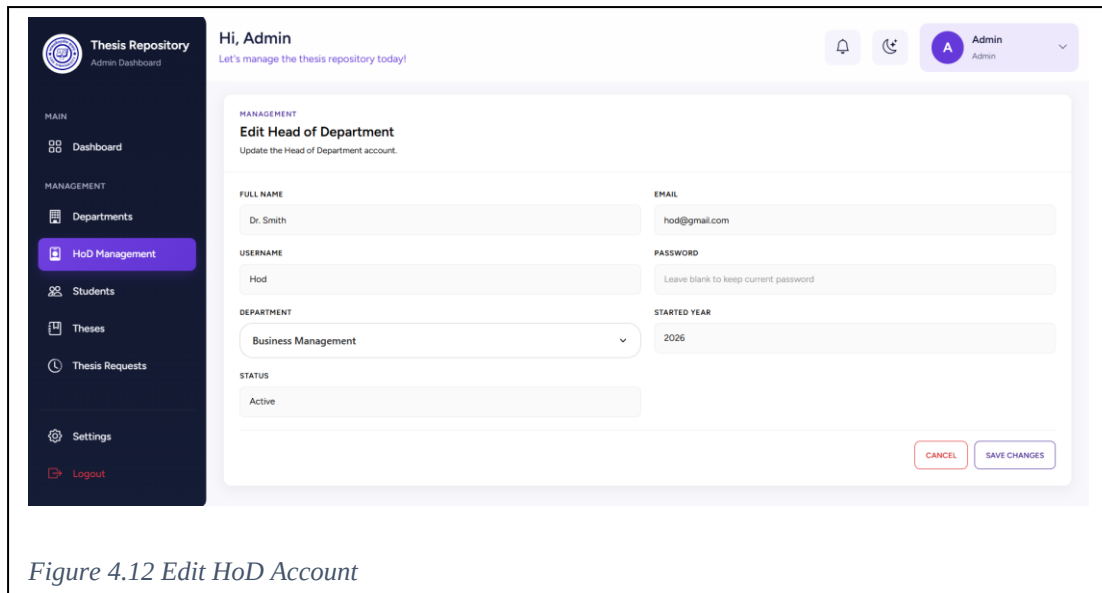


Figure 4.12 Edit HoD Account

4.2.3.4 Student Upload Permission Management

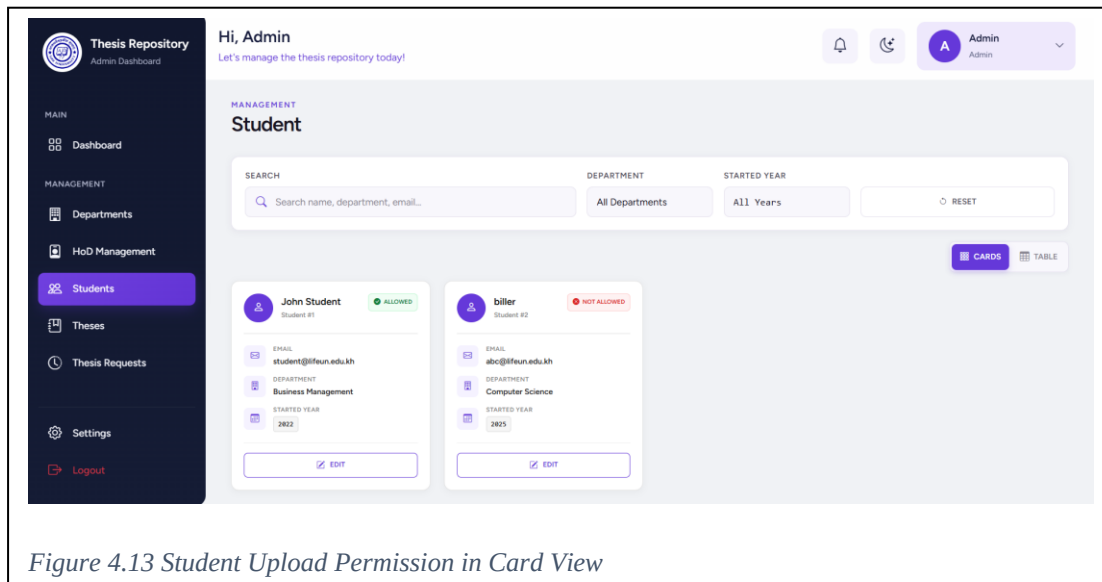


Figure 4.13 Student Upload Permission in Card View

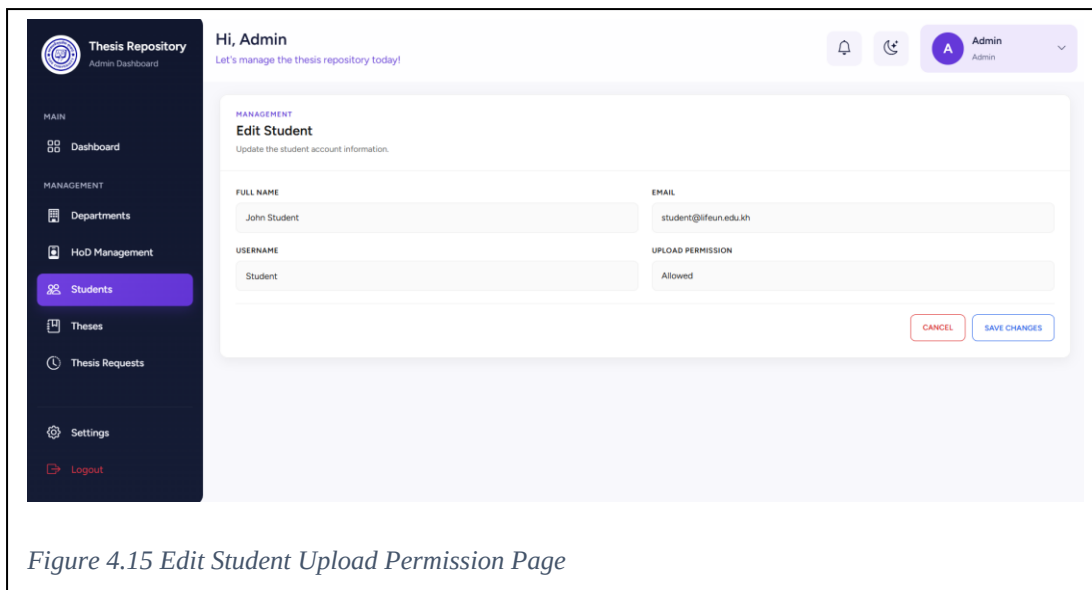


Figure 4.15 Edit Student Upload Permission Page

4.2.3.5 Thesis Management

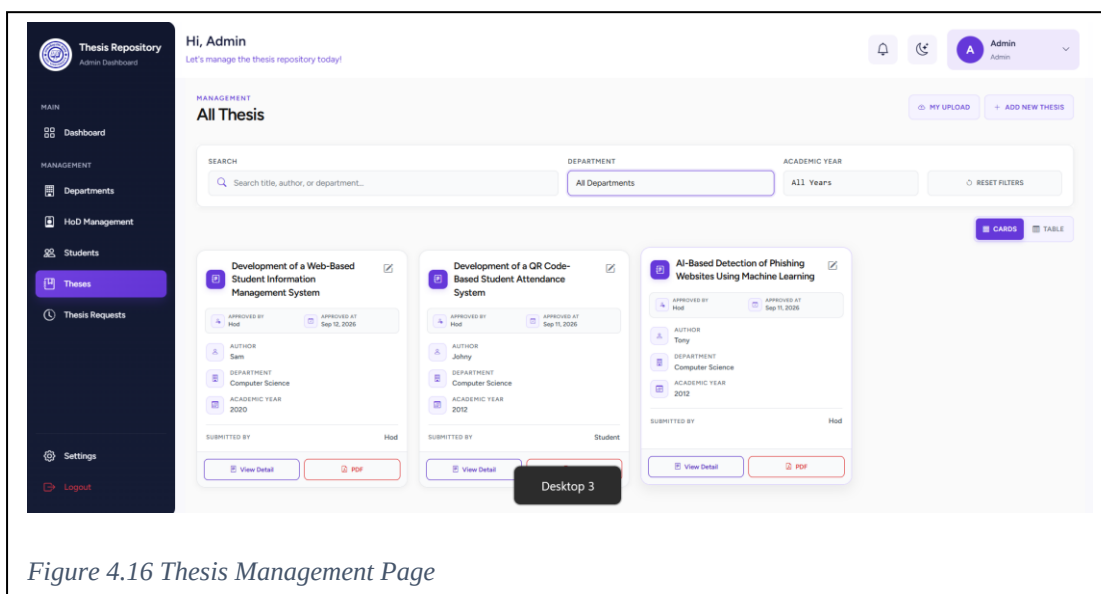


Figure 4.16 Thesis Management Page

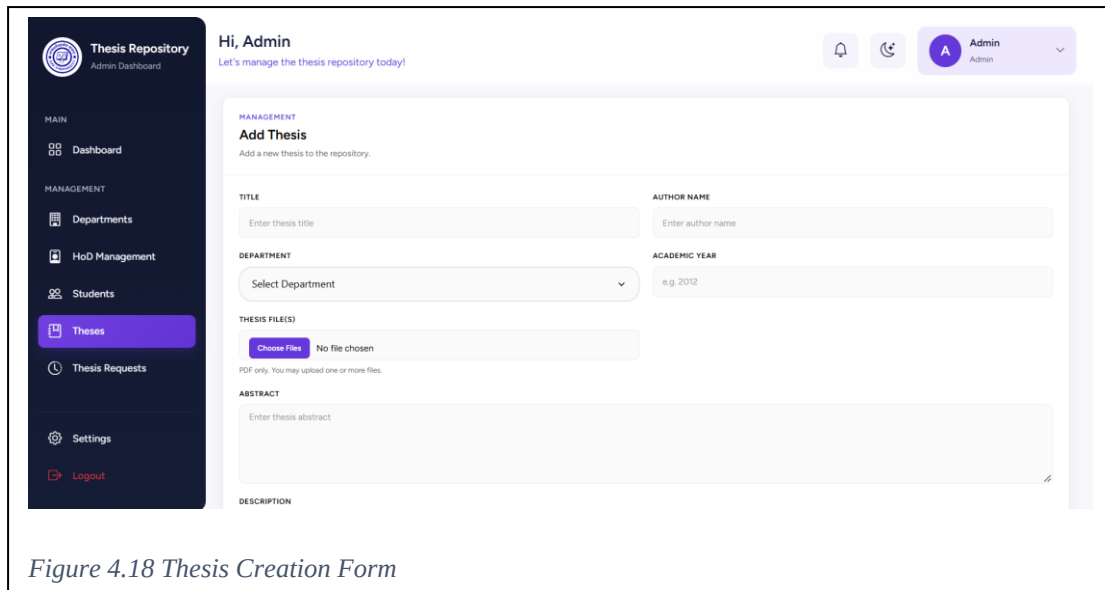


Figure 4.18 Thesis Creation Form

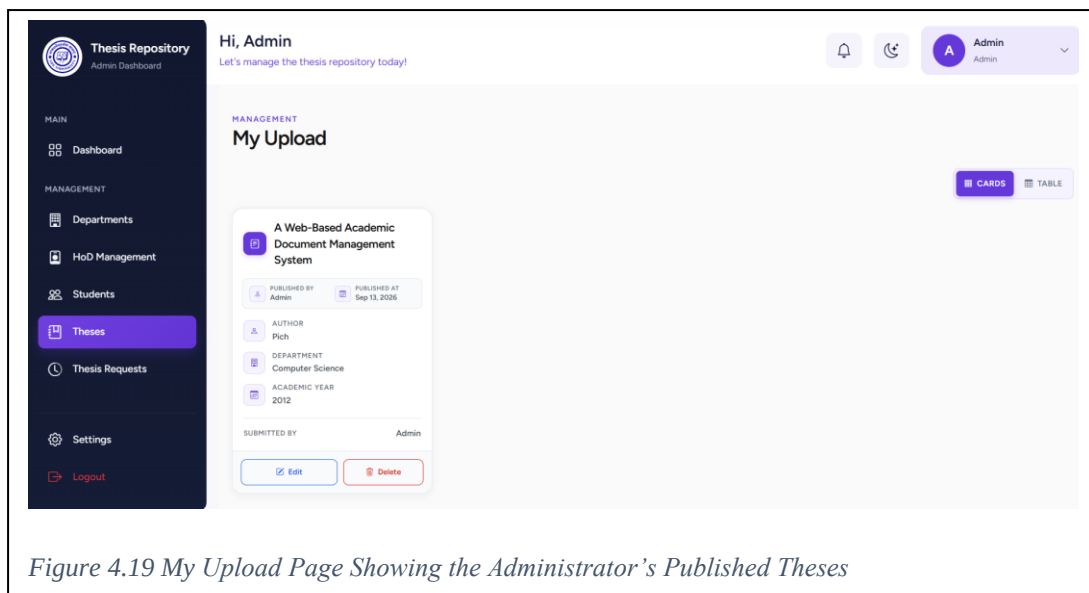


Figure 4.19 My Upload Page Showing the Administrator's Published Theses

4.2.3.6 Thesis Publication Request

The Thesis Publication Request page allows students with upload permission to submit a request to publish their completed thesis. The page displays the student's submitted thesis publication requests. If the student has not submitted any thesis publication requests, the page displays an empty state indicating that no thesis publication request is available.

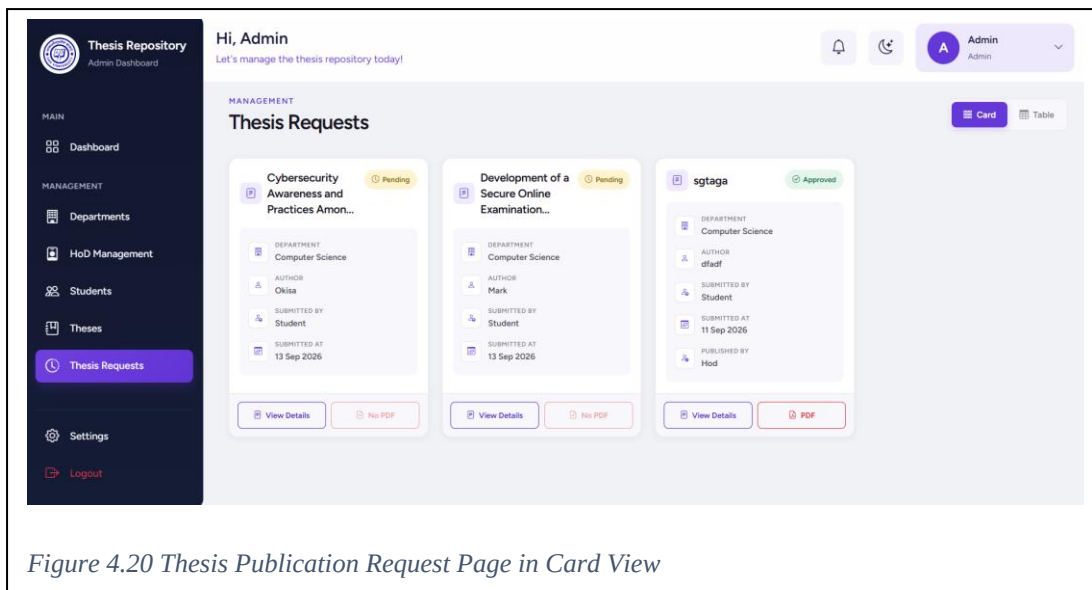


Figure 4.20 Thesis Publication Request Page in Card View

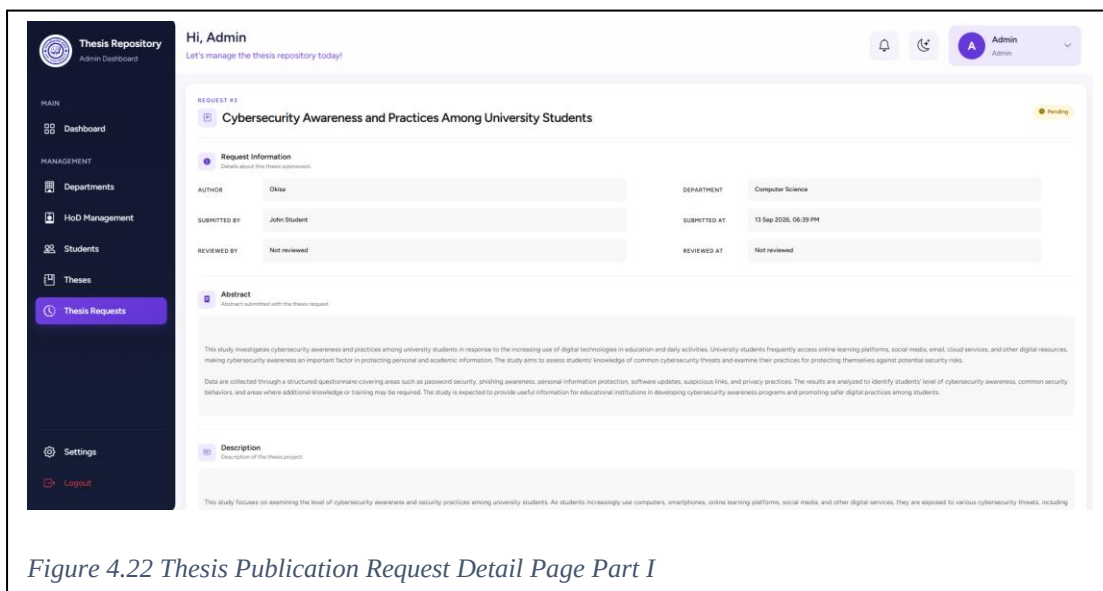
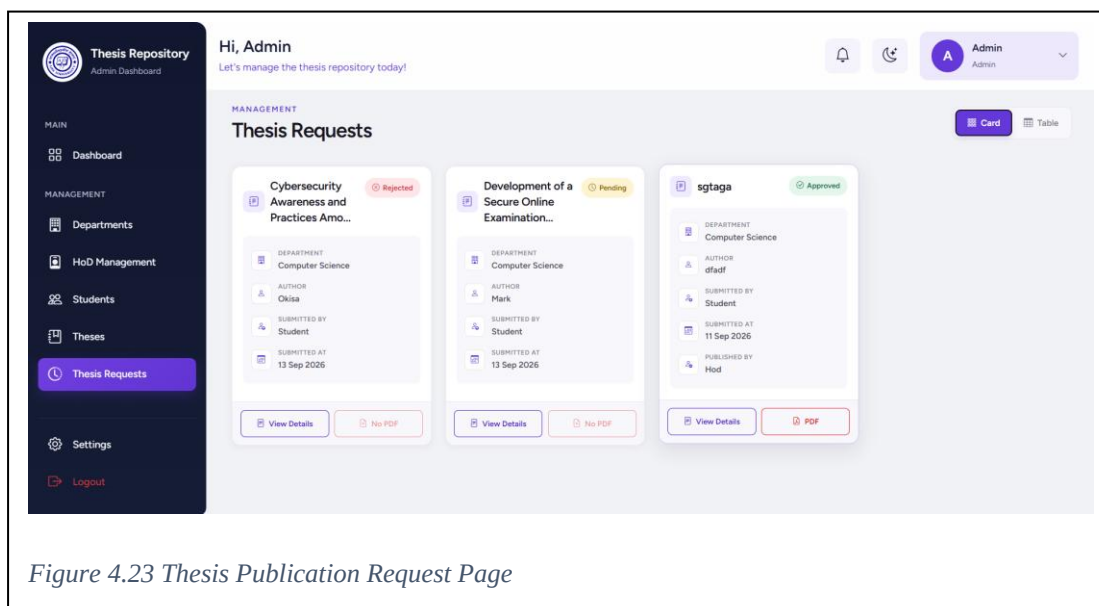


Figure 4.22 Thesis Publication Request Detail Page Part I



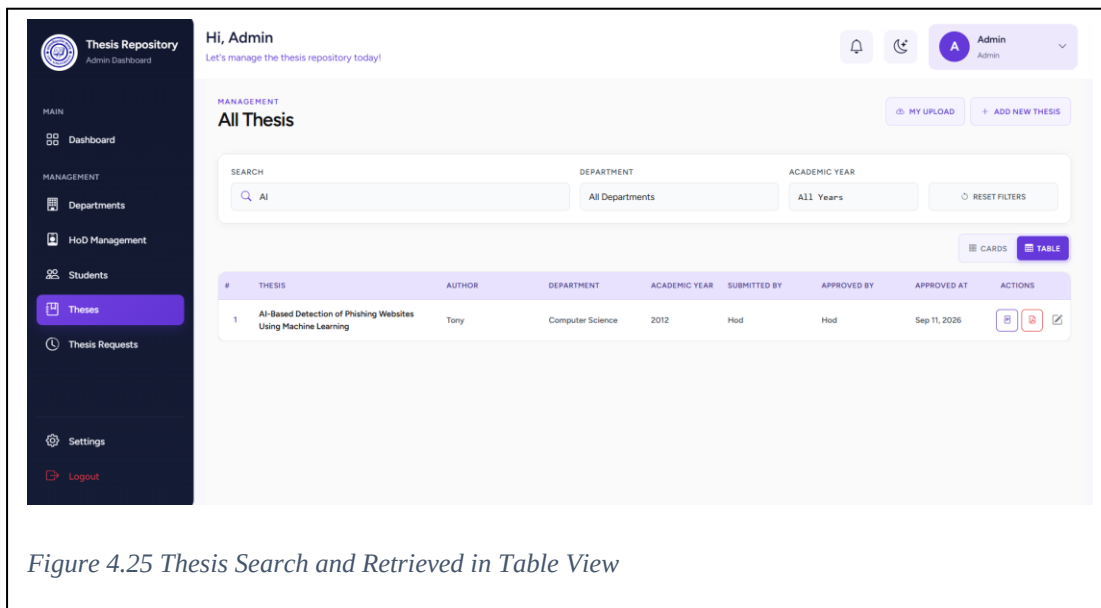
The Administrator can approve the request to publish the thesis in the repository. If the request is rejected, the Administrator must provide a comment explaining the reason for rejection. Then, the request status is changed from Pending to either Approved or Rejected based on the Administrator's decision.



4.2.3.7 Thesis Search and Retrieval

The Thesis Search and Retrieval page allows the Administrator to search for published theses in the repository. Administrator can search by thesis title, author, department, academic year, or keyword. The search results display the matching theses,

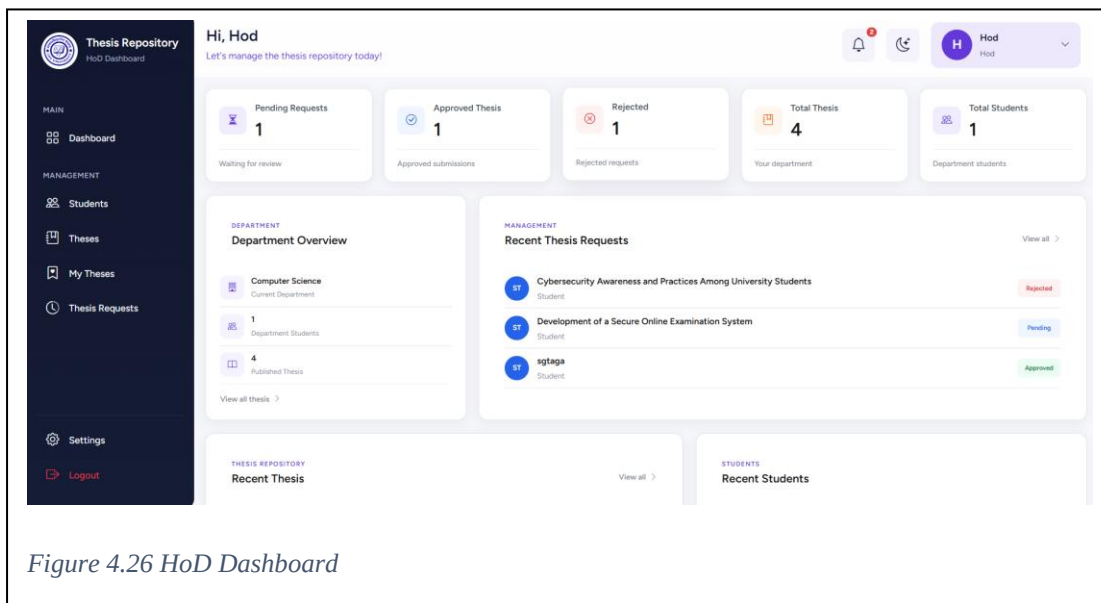
and the Administrator can select a thesis to view its details and access the thesis file.



4.2.4 Head of Department Module

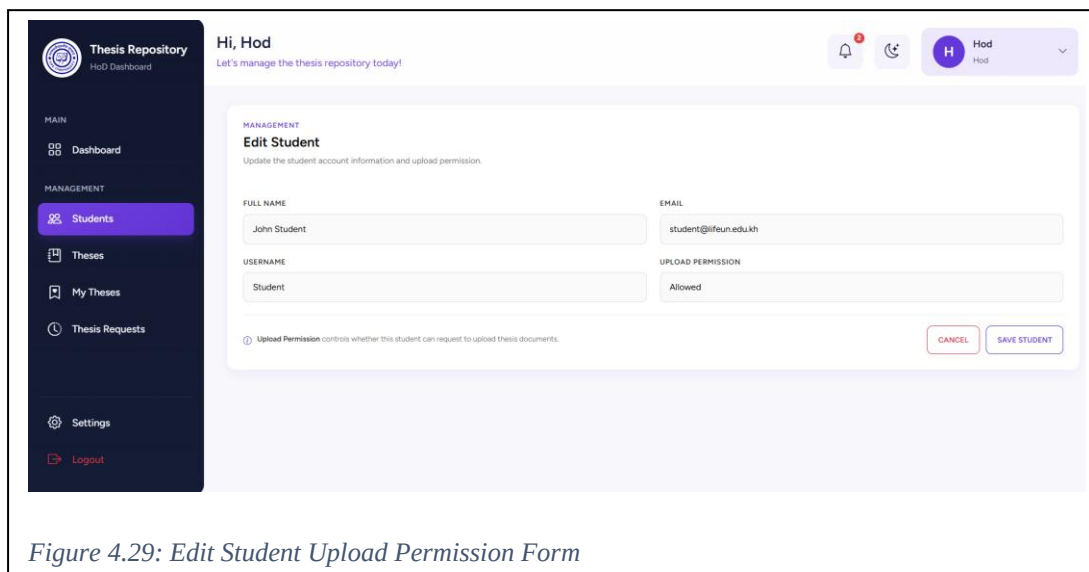
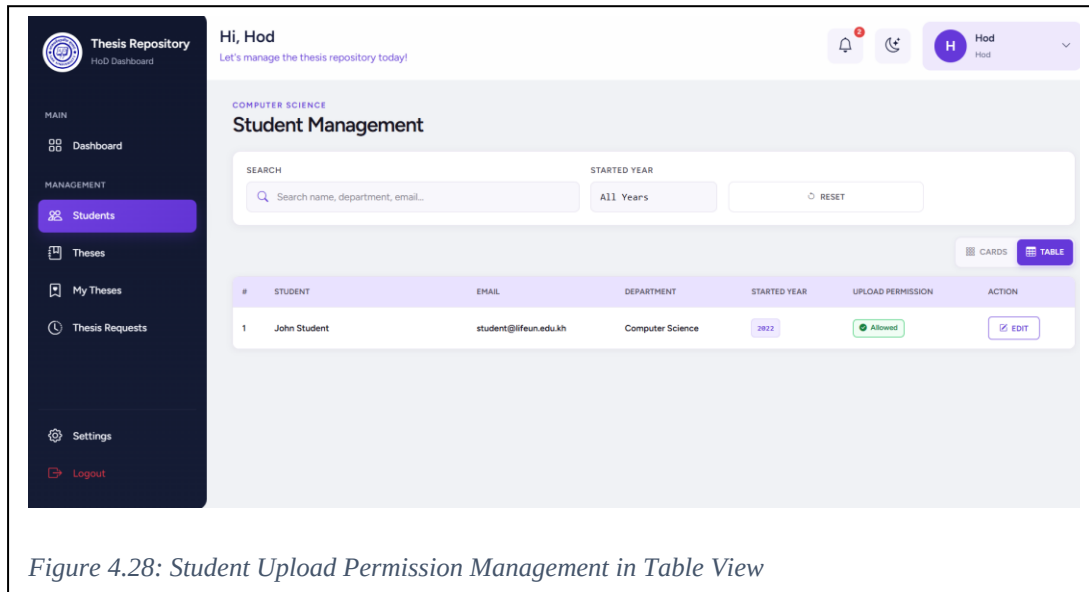
Before accessing the Head of Department Dashboard, the Head of Department (HoD) must have an account created by the Administrator. This ensures that only authorized HoD accounts can access the system. After logging in successfully, the HoD is directed to the Head of Department Dashboard.

4.2.4.1 Head of Department (HoD) Dashboard



4.2.4.2 Student Upload Permission Management

This page is similar to the Administrator's Student Upload Permission Management page. However, the Head of Department (HoD) can only grant or revoke upload permission for students within their own department and cannot manage students from other departments.



4.2.4.3 Thesis Management

This page displays published theses from all departments. The HoD can view the thesis details, read the thesis, and download the published thesis.

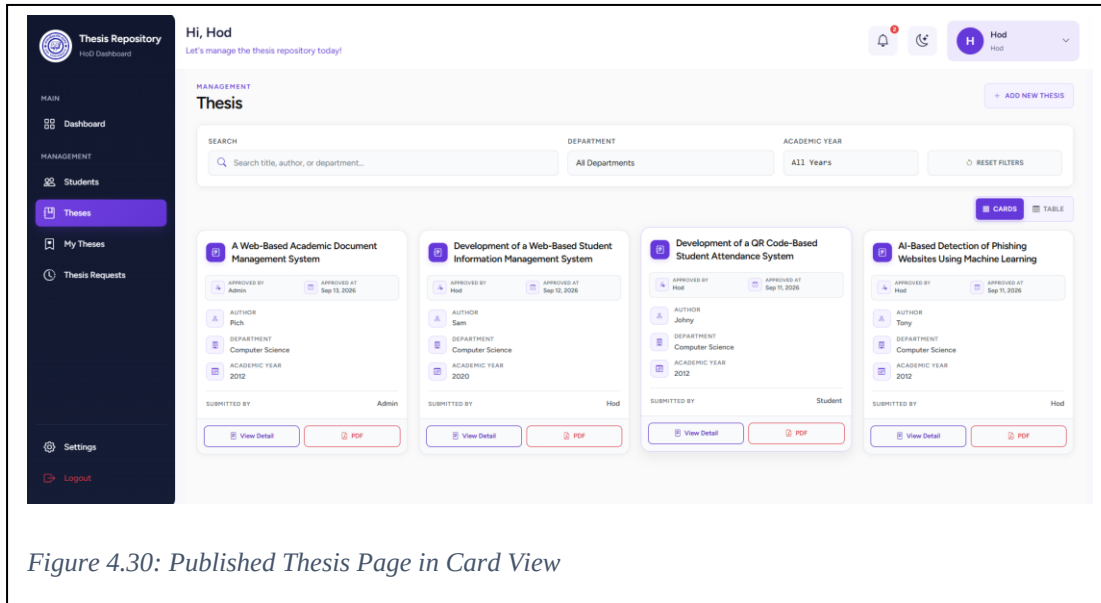


Figure 4.30: Published Thesis Page in Card View

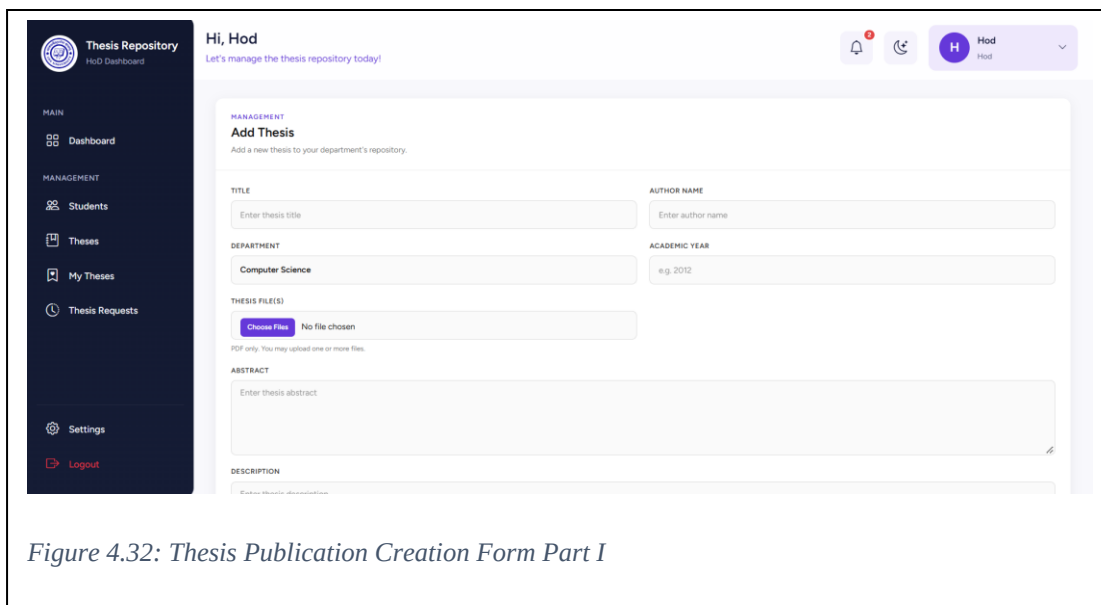
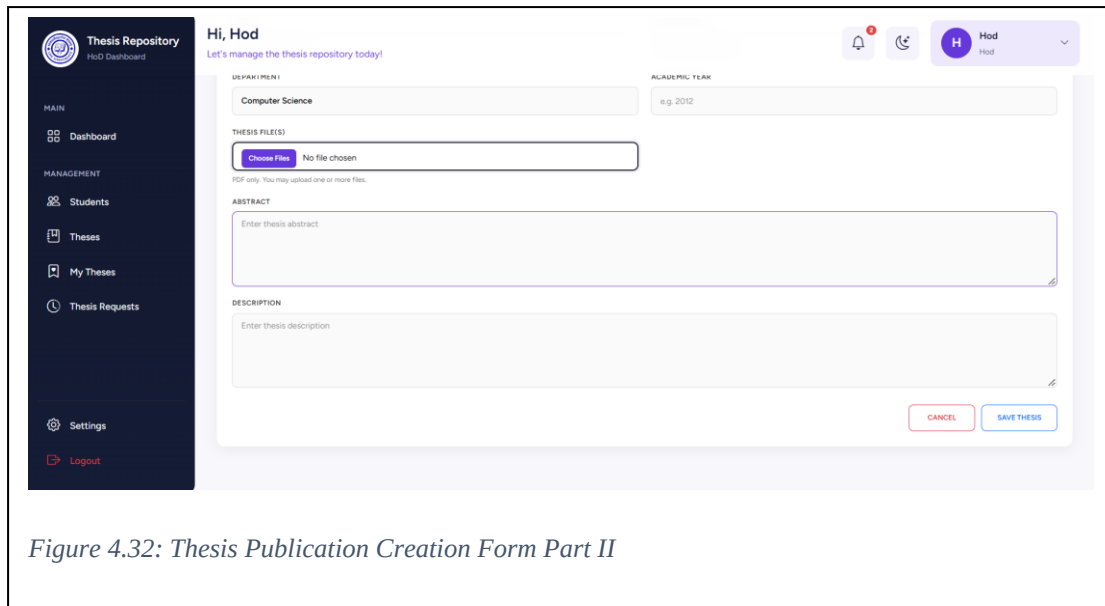
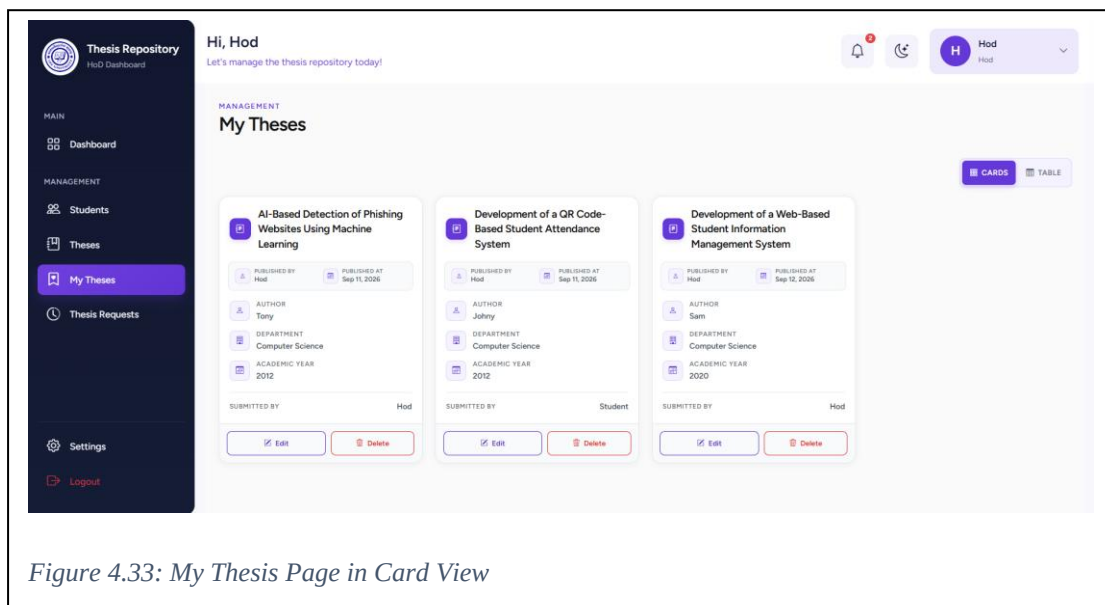


Figure 4.32: Thesis Publication Creation Form Part I



4.2.4.4 My Thesis Management

After a thesis is published, it will be displayed on My Thesis Management page. The HoD can view and manage theses published by the HoD. Other HoDs can only view the published theses on the Thesis Management Page and cannot manage them, except for the administrator, who can manage all published theses.



4.2.4.5 Thesis Publication Request

As mentioned earlier, the HoD has access to manage thesis publication requests from students within the respective department. The HoD can view, approve, or

reject thesis publication requests submitted by students in the HoD's department. The process is the same as the thesis publication request process for the administrator.

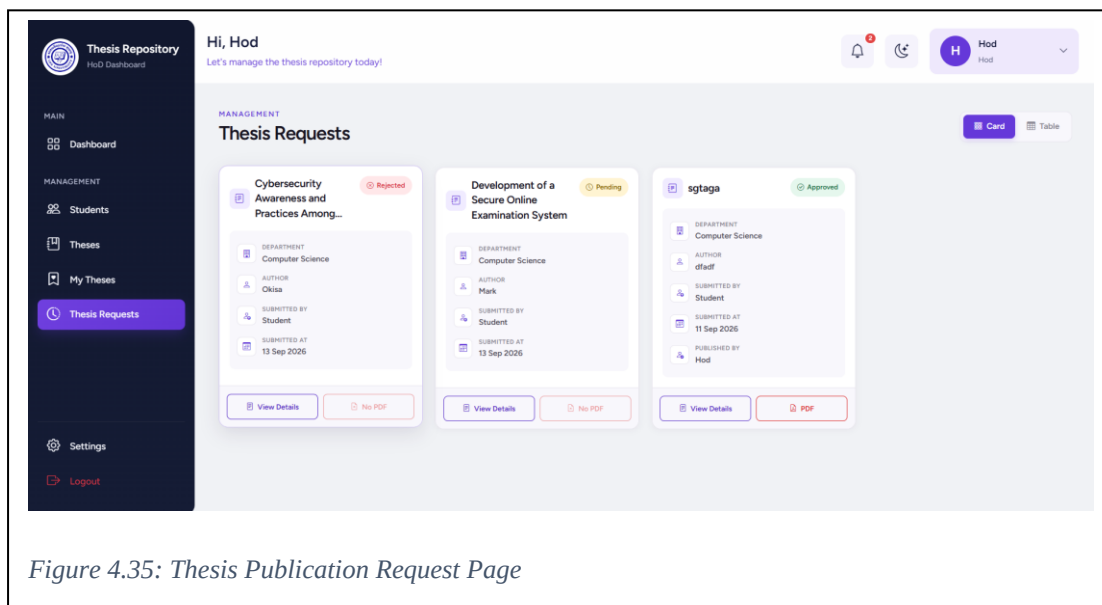
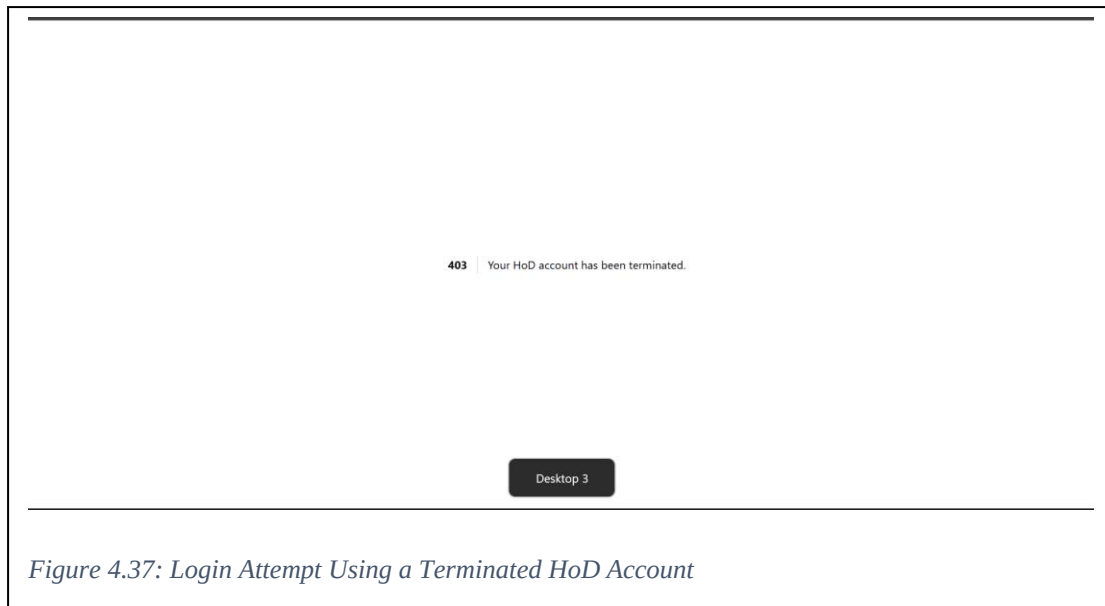


Figure 4.35: Thesis Publication Request Page

4.2.4.6 Terminate Account of HoD

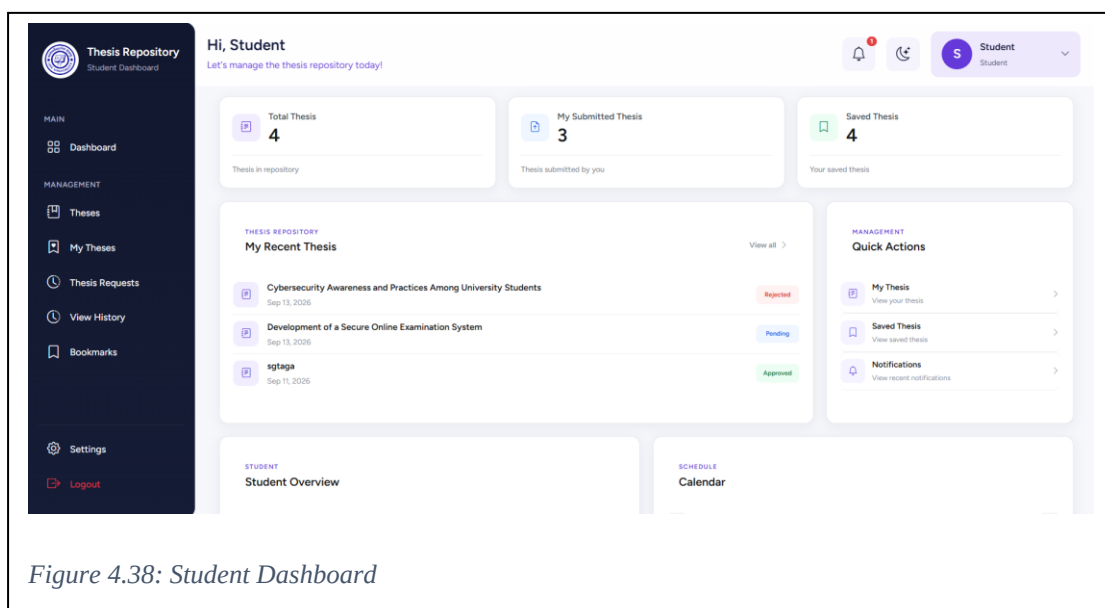
As discussed earlier, the HoD account must be created by the administrator before the HoD can log in to the HoD Dashboard. The account has two statuses: Active and Terminated. An HoD can log in to the system only when the account is in Active status. When an HoD leaves the university, the administrator changes the account status to Terminated through the HoD Management page in the Admin Module instead of deleting the account. The account is not deleted because the HoD may be associated with existing system records, such as published theses and other related information. Deleting the account could affect the integrity of these records. After the account status is changed to Terminated, the former HoD can no longer log in to the system, while the account and its associated information remain preserved in the system.



4.2.5 Student Module

As mentioned in Section 4.3.2.1, User Registration, the system identifies a Life University student when a school email address (e.g., abc@lifeun.edu.kh) is entered during registration. After the email address is successfully verified, the student can log in to the system and will be directed to the Student Dashboard.

4.3.5.1 Student Dashboard



4.2.5.2 Thesis Browsing, Viewing, Reading and Downloading

Students can browse, view, search, read, and download published theses through the system. The system allows students to search for theses using different criteria, including author name, thesis title, department, and academic year.

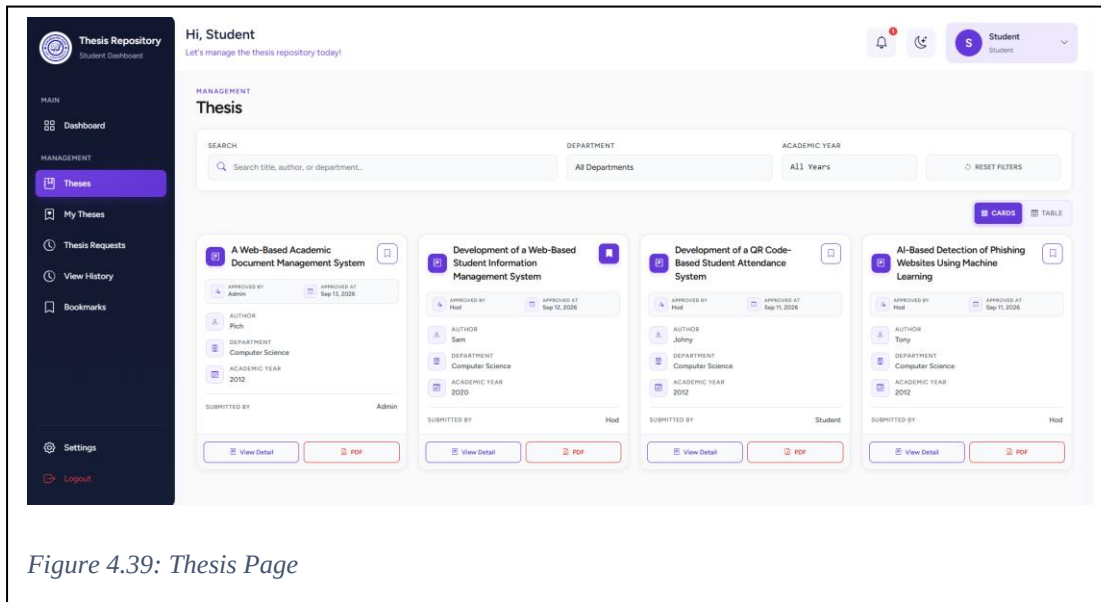


Figure 4.39: Thesis Page

4.2.5.3 Student with Upload Permission

A student with upload permission can submit a thesis publication request after the thesis has been completed and passed the final defense. The student can also view the status of the submitted request.

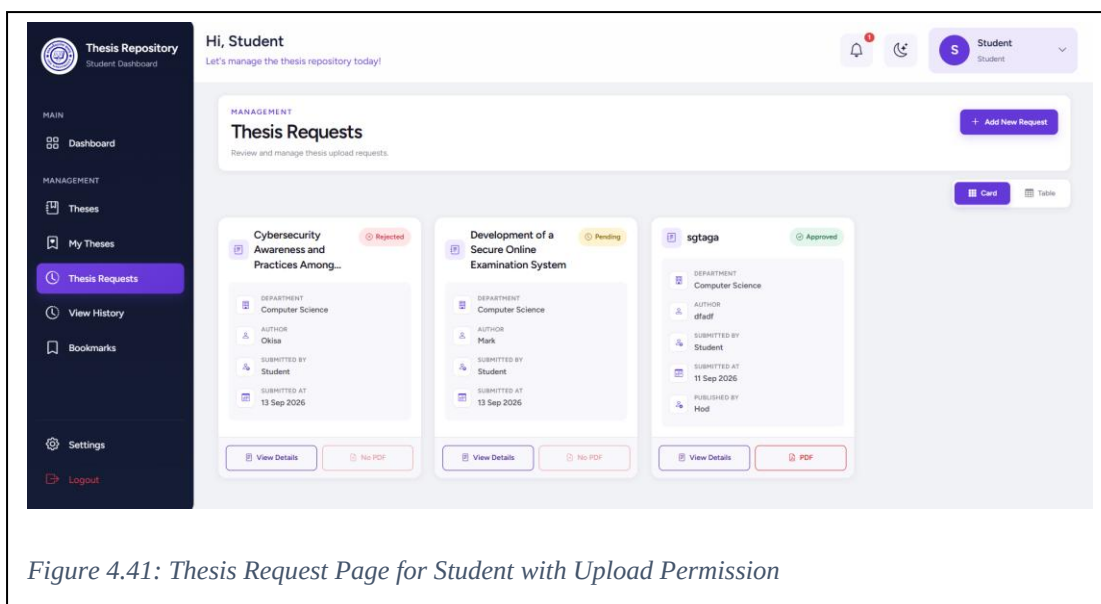


Figure 4.41: Thesis Request Page for Student with Upload Permission

4.2.5.3 Student without Upload Permission

The following figure appear when a student without upload permission accesses the Thesis Request section from the sidebar. The page displays all the requirements for obtaining upload permission before sending a thesis publication request.

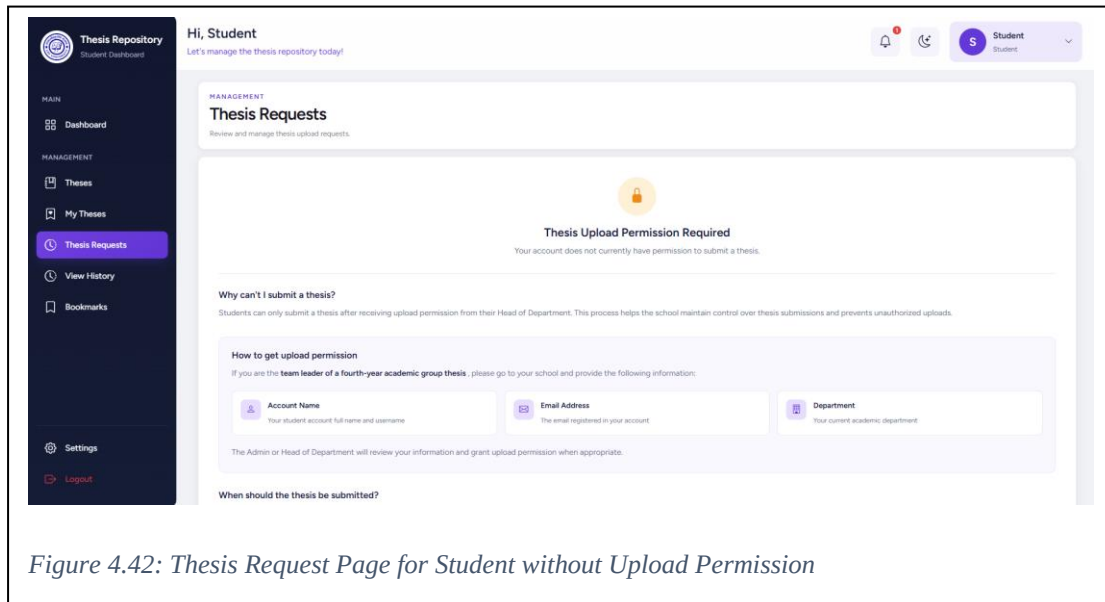


Figure 4.42: Thesis Request Page for Student without Upload Permission

4.2.5.4 Thesis Request Status

The Thesis Request Status page allows students to view the status of submitted thesis publication requests, including Pending, Approved, or Rejected.

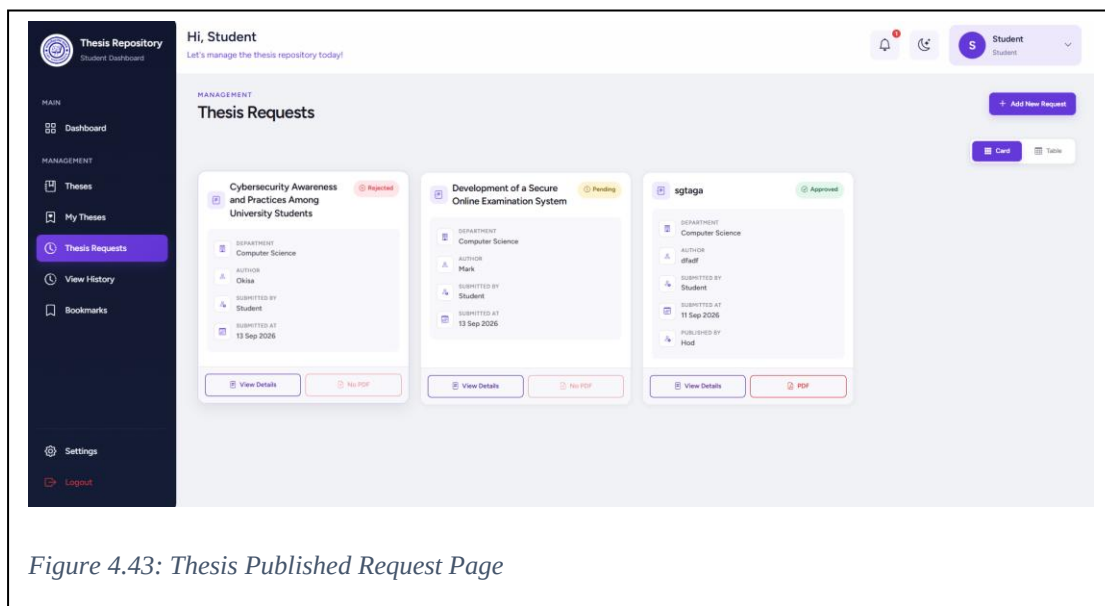


Figure 4.43: Thesis Published Request Page

Figure 4.45: Thesis Published Request Creation Form

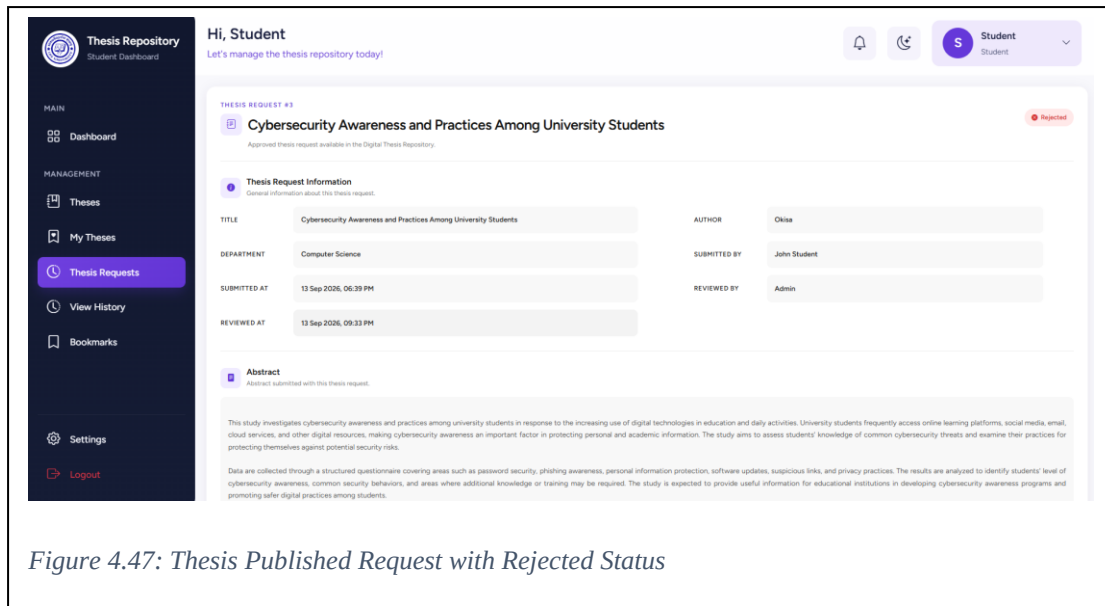
Thesis Request Information	
TITLE	Development of a Secure Online Examination System
DEPARTMENT	Computer Science
SUBMITTED AT	13 Sep 2026, 06:28 PM
REVIEWED AT	15 Sep 2026, 04:33 PM

Abstract
Abstract submitted with this thesis request.

This study presents the design and development of a secure online examination system for improving the efficiency, reliability, and security of examination processes in educational institutions. Conventional examination methods involve paper-based materials and manual procedures that can require considerable time and resources and may increase the risk of errors, unauthorized access, and loss of examination records. To address these challenges, the proposed system provides a centralized web-based platform for creating examinations, managing questions, scheduling assessments, conducting examinations, and generating examination results.

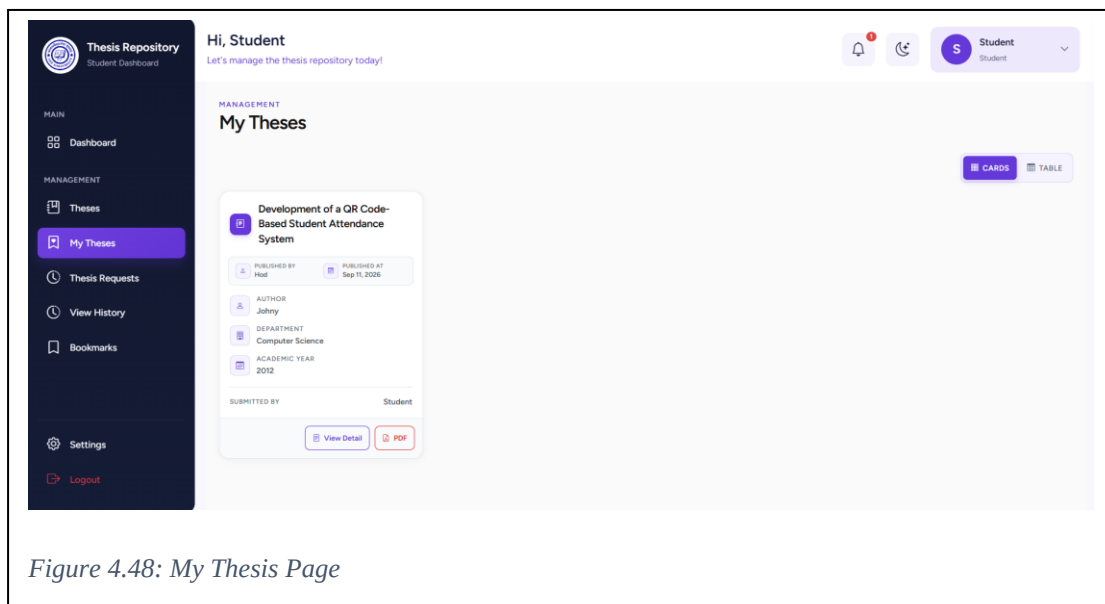
The system implements role-based access control to restrict access to examination functions and sensitive information according to user responsibilities. Automatic scoring and result generation are also provided to reduce manual calculation and improve the accuracy of assessment results. The system aims to provide students with a convenient examination environment while allowing administrators and instructors to efficiently manage examination activities. The proposed system demonstrates how secure web technologies can be applied to create a more organized, accessible, and reliable online examination process.

Figure 4.46: Thesis Published Request with Approved Status



4.2.5.5 My Thesis Page

After the thesis publication request is approved, the thesis will be published and displayed on the My Thesis page of the student who submitted the request.



4.2.6 Guest User Module

The system identifies a guest user through a general email address. A guest user can browse, search, view, read, and download published theses. Guest users do not have permission to submit thesis publication requests or upload theses.

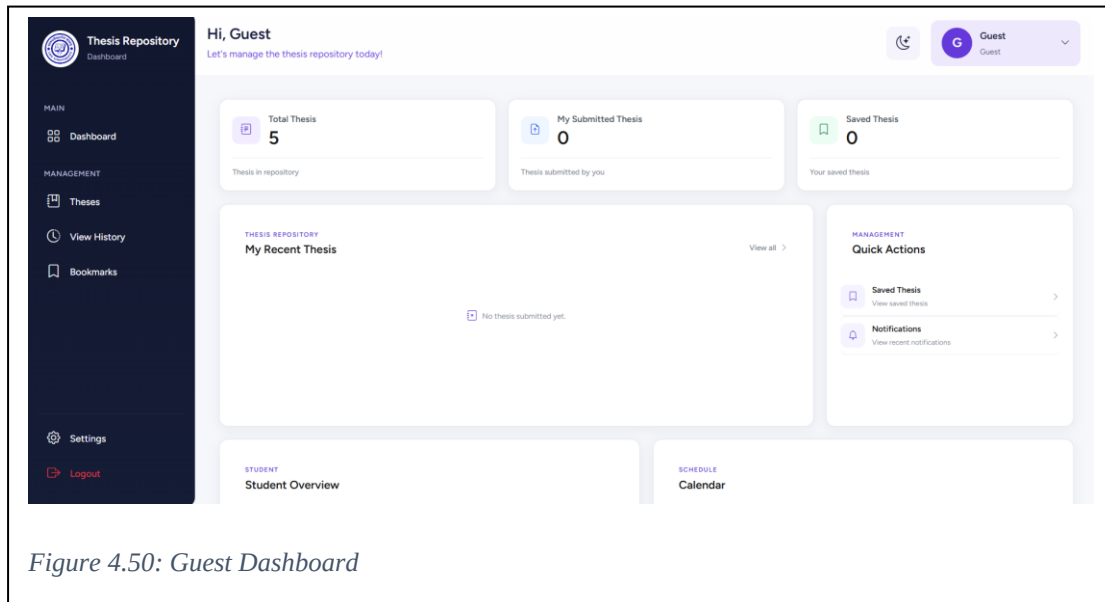


Figure 4.50: Guest Dashboard

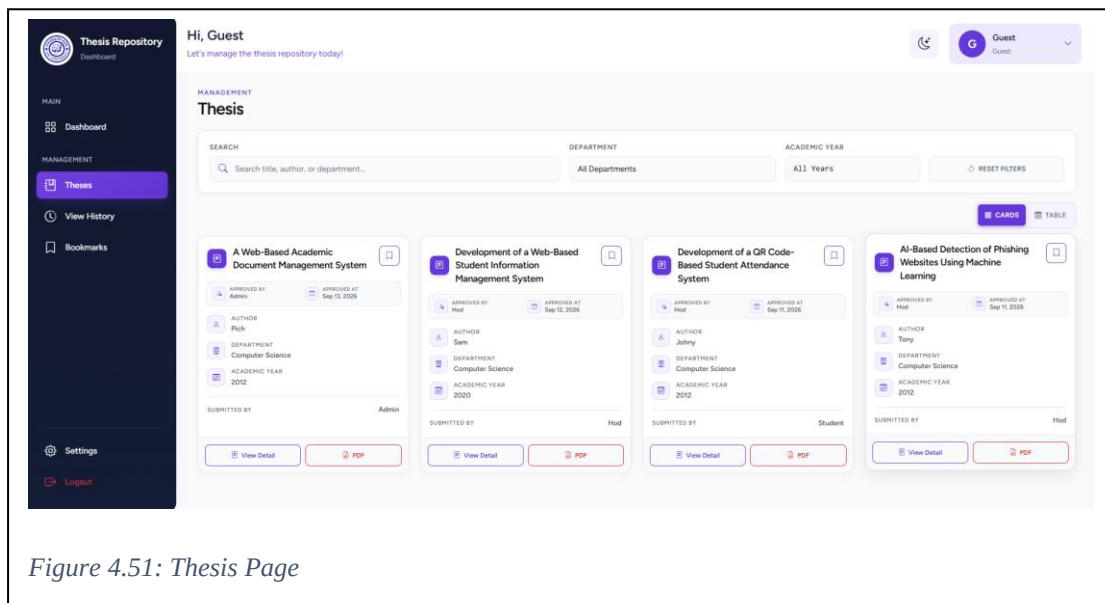


Figure 4.51: Thesis Page

4.2.7 Notifications

The notification feature informs users about important system activities.

Administrator and HoDs receive notifications about thesis publication requests from students, and students receive notifications about the status of submitted thesis publication requests.

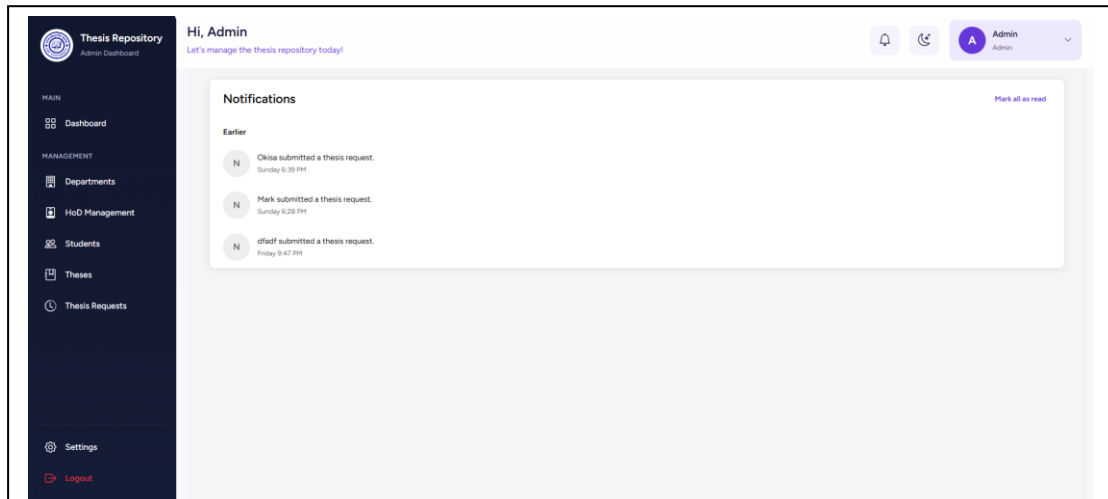


Figure 4.53 Notification of a Student's Thesis Publication Request in the Admin Module

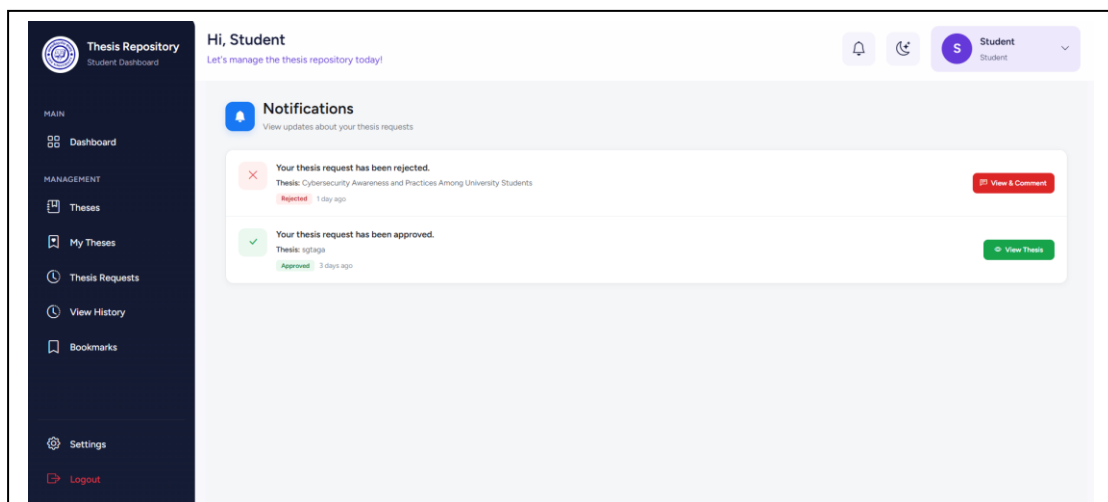


Figure 4.54 Notification of an Approved or Rejected Thesis Publication Request in the Student Module

4.2.8 Dark Mode

The system provides a Dark Mode feature that can be applied to all pages. Users can switch between Light Mode and Dark Mode to improve readability and provide a more comfortable viewing experience while using the system.

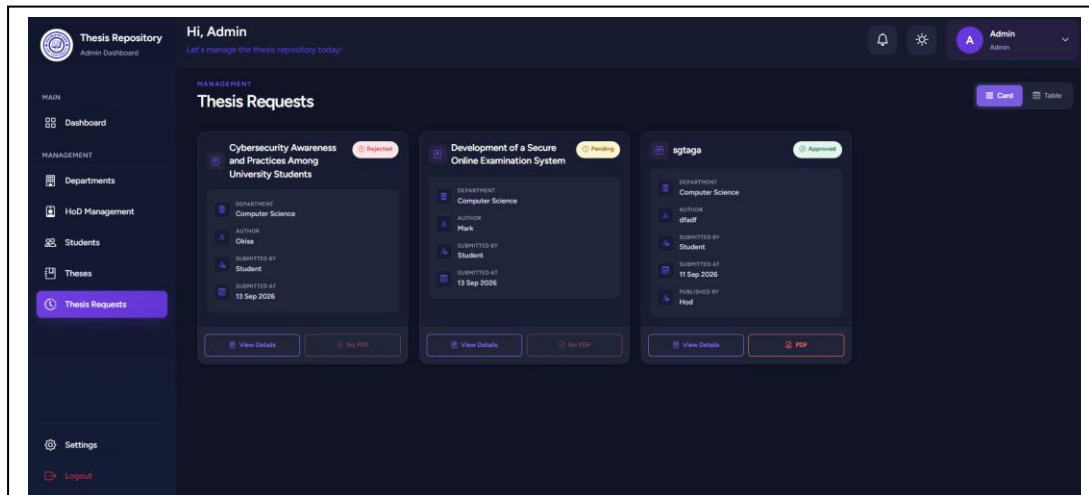


Figure 4.55 Thesis Request of Admin in Dark Mode

4.2.9 View History and Saved Bookmarks

The system provides View History and Saved Bookmarks features for students and guest users. View History records previously viewed theses for quick access, while Saved Bookmarks allows users to save published theses for future reference.

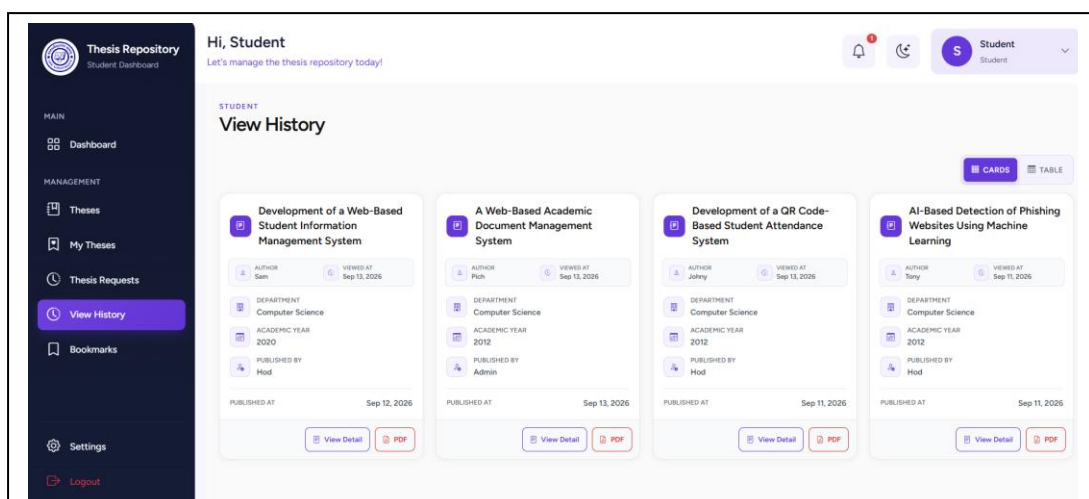
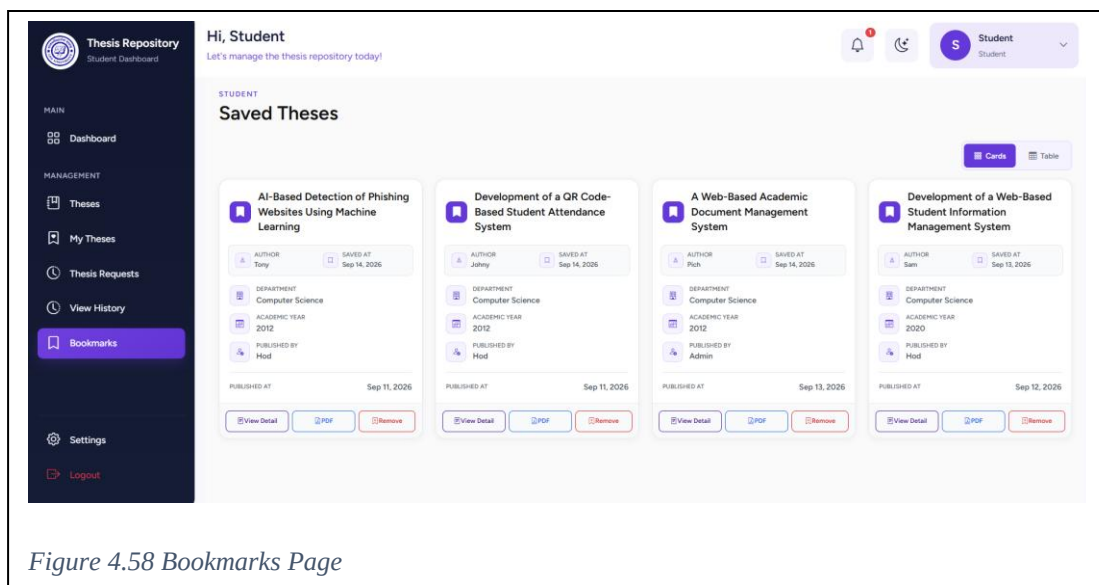
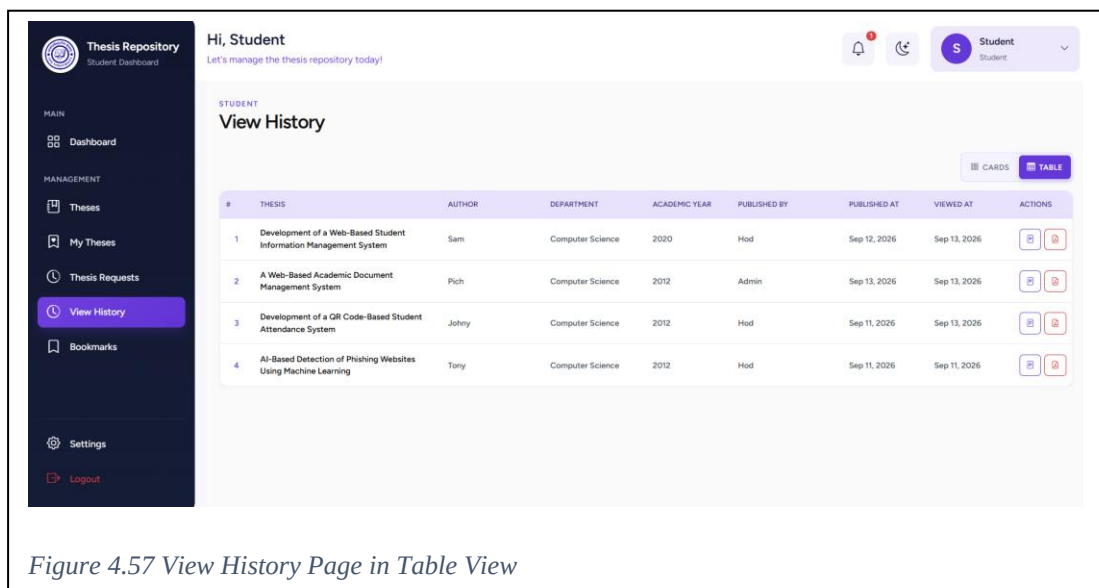


Figure 4.56 View History Page in Card View



4.3 System Testing

System testing was conducted to verify that the DTRP met its defined requirements. Black-box testing was used to evaluate system functions, role-based access, and email verification by checking inputs against expected outputs. The testing aimed to identify errors and ensure correct system performance.

4.3.1 Testing Approached

The system was tested using the Black-box Testing approach to verify its functions without examining the source code. Testing covered Functional Testing, Role-Based Access Testing, and Email Verification Testing. Each test case was evaluated by comparing the actual result with the expected result and recorded as Pass or Fail.

4.3.2 Functional Testing

Functional testing was performed to verify that the main functions of the DTRP operated according to the system requirements.

No.	Test Case	Expected Result	Result
1	Student Registration	A Life University email creates student account and sends a verification email.	Pass
2	Guest Registration	An ordinary email creates a guest account.	Pass
3	Login	Users can log in using valid credentials.	Pass
4	Logout	Users can successfully log out of the system.	Pass
5	Thesis Search	The system displays these matching the search keyword.	Pass
6	Thesis Filtering	The system filters theses according to the selected criteria.	Pass
7	View Thesis Detail	The system displays the selected the information.	Pass
8	View PDF	The selected thesis PDF can be opened and viewed.	Pass
9	Download PDF	The selected thesis PDF can be downloaded.	Pass
10	Thesis Publication Request	An authorized student can submit a thesis publication request.	Pass
11	Approve Thesis Request	Admin or HoD can approve a thesis publication request	Pass
12	Reject Thesis Request	Admin or HoD can reject a thesis publication request.	Pass
13	Thesis Management	Authorized users can manage thesis records.	Pass
14	Student Management	Admin or HoD can manage student records.	Pass
15	HoD Management	Admin can manage HoD Accounts.	Pass
16	Notification	The system displays notifications for relevant activities.	Pass
17	Bookmark	Users can bookmark and access saved theses.	Pass
18	History	Users can view their thesis viewing history.	Pass

Table 4.1 Role-Based Access Testing of the DTRP Using Black-Box Testing

The results confirmed that the system restricted access according to the assigned user role and prevented unauthorized access to restricted functions.

4.3.3 Email Verification Testing

Email verification testing was conducted to verify that the email verification process worked correctly for student registration. The testing focused on

registration using a Life University email address, receiving the verification email, and accessing the system after verification.

No.	Test Case	Expected Result	Result
1	Student Email Registration	A user registering with a Life University email is identified as a student.	Pass
2	Verification Email Sent	The system sends a verification email to registered email address.	Pass
3	Verification Link	The user can verify the email by clicking the verification link.	Pass
4	Unverified Account Protection	An unverified student cannot access functions that require email verification.	Pass
5	Verified Account Access	A verified student can access the student functions	Pass
6	Guest Email Registration	A user registering with an ordinary email is identified as a guest.	Pass

Table 4.2 Email Verification Testing of the DTRP Using Black-Box Testing

The results confirmed that email verification worked as expected, ensuring that student accounts were verified before accessing student functions.

4.3.4 Testing Results

The overall testing results are summarized below.

Testing Areas	Test Case	Passed	Failed	Pass Rate
Functional Testing	18	18	0	100%
Role Based-Access Testing	18	18	0	100%
Email Verification Testing	6	6	0	100%
Total	42	42	0	100%

Table 4.3 Overall Testing Results of the DTRP

The black-box testing consisted of 42 test cases covering functional testing, role-based access testing, and email verification. All 42 test cases passed, while no test cases failed. Therefore, the overall confirming that the system functions, role-based access, and email verification worked as expected.

4.4 Performance Testing

Performance testing was conducted using k6 to evaluate the system under concurrent access by different user roles. The test simulated 100 Students, 20 HoDs, and 10 Admins, for a total of 130 virtual users running concurrently for 30 seconds. The test

completed 720 iterations and generated 7,080 HTTP requests. The average response time was 51.84 ms, with a 95th percentile response time of 194.80 ms and a maximum response time of 504.18 ms. The HTTP request failure rate was 0%, and all checks passed successfully. These results show that the system handled the tested workload successfully in the local development environment. These results show that the system handled the tested workload successfully in the local development environment.

Metric	Result
Total VUs	130
Student/ HoD/ Admin	100 / 20 / 10
Duration	30 sec
Iterations	720
HTTP requests	7,080
Average response time	51.84 ms
P95 response time	194.80 ms
Maximum response time	504.81 ms
HTTP failure rate	0 %
Checks passed	100%

4.5 Task Performance Evaluation

Task performance evaluation was conducted to measure the time required to complete four common tasks in the Digital Thesis Repository Platform: searching for a thesis, applying a filter, opening a thesis PDF, and downloading a thesis PDF. Each task was performed five times, and the average completion time was calculated from the five trials.

Task	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average
Search for a thesis	4.36s	5.70s	3.00s	4.03s	4.03s	4.10s
Apply a filter	4.27s	4.26s	3.60s	3.62s	3.23s	3.80s
Open a PDF	10.41s	4.53s	3.50s	5.12s	3.73s	5.46s

Download a PDF	7.09s	5.64s	3.00s	5.21s	4.62s	5.11s
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Table 4.4 Overall Testing Results of the DTRP

The average completion time was calculated using the following formula:

$$\text{Average Time} = \frac{\text{Trial 1} + \text{Trial 2} + \text{Trial 3} + \text{Trial 4} + \text{Trail 5}}{5}$$

Therefore, the task performance evaluation shows that the proposed Digital Thesis Repository Platform can complete common thesis-related tasks within a short period of time. The average time to search for a thesis was 4.10 seconds, while applying a filter required an average of 3.80 seconds. Opening a thesis PDF took an average of 5.46 seconds, and downloading a PDF took an average of 5.11 seconds. These results demonstrate that the system can perform common thesis access tasks efficiently. However, these measurements represent the performance of the proposed system only and do not indicate a percentage improvement over the manual system because reliable manual system timing data was not available.

4.7 User Evaluation

User evaluation was conducted using a five-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree). A total of 13 students participated in the evaluation. The evaluation assessed the system's usability, accessibility, search, functionality, efficiently, features, and overall satisfaction.

The evaluation results were analyzed using the percentage provided by the Google Forms response charts. For each question, the percentages of Agree and Strongly Agree were combined to determine the positive response rate. The following table presents the positive response percentage for each evaluation question. The corresponding pie charts of the evaluation results are presented on page 91, as shown below:

Question	Agree (%)	Strong Agree (%)
Q1. System is easy to use	84.6	7.7
Q2. Help users find useful information	76.9	15.4
Q3. Search and filter function are useful	92.3	7.7
Q4. Thesis history feature is easy to use	30.8	0
Q5. Viewing and downloading PDFs is easy	76.9	7.7
Q6. Provides functions needed for my role	41.7	0
Q7. Notification feature is useful	69.2	0
Q8. Makes thesis management and access easier	69.2	23.1
Q9. Bookmark feature is useful	36.4	0
Q10. Uploading a thesis is easy	46.2	0
Q11. Helps save time	76.9	23.1
Q12. HoD/Admin feedback is useful	61.5	7.7
Q13. Overall performance satisfaction	69.2	15.4
Q14. Would recommend the system	76.9	23.1
Total	908.7	130.9

The overall positive response (OPR) was calculated by adding the sum of percentages of Agree (SPA) and sum of percentage of Strongly Agree (SPSA) for all questions and dividing the total by the number of questions.

$$OPR (\%) = \frac{(SPA (\%) + SPSA (\%))}{Total\ Number\ of\ Questions}$$

$$\text{Using the results from the 14 evaluation questions: } OPR = \frac{908.7 + 130.9}{14} = 74.26\%$$

Therefore, the evaluation achieved a positive response rate of 74.26%. This result indicates the level of improvement reported by users after using the DTRP compared with the previous manual system.

4.8 Discussion

The testing results show that the Digital Thesis Repository Platform (DTRP) met its main functional and access-control requirements. A total of 42 black-box test cases were conducted, and all 42 passed, giving an overall pass rate of 100%.

Functional testing confirmed that the main features, such as authentication, thesis

management, requests, searching, and file access, worked as expected. Role-based testing confirmed that each user could only access functions allowed for their role. Email verification also worked correctly for Life University student accounts.

Performance testing using k6 also showed that the system successfully handled a combined workload of 130 virtual users, including 100 Students, 20 HoDs, and 10 Admins. During the 30 second test, 7,080 HTTP requests were completed with a 0% failure rate. The average response time was 51.84 ms, while the 95th percentile response time was 194.80 ms. Task performance testing covered four common tasks, with 5 trials each. Average completion time was 3.80-5.46 seconds.

Overall, the results indicate that the system can support the management, storage, searching, and access of completed theses. The system also demonstrated stable performance under the tested workload. However, the testing was conducted in a controlled environment, so further testing with more users would be useful to evaluate real-world performance and usability.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The project successfully developed a Digital Thesis Repository Platform for Life University to improve the storage, management, search, and accessibility of thesis documents. The system provides a centralized platform with role-based access for Administrators, Heads of Department, Students, and Guests.

The platform includes thesis management, search and filtering, publication requests, PDF viewing and downloading, bookmarks, notifications, history, and email verification. Role-based access control ensures that users can access functions according to their responsibilities.

System testing using black-box testing showed that all 43 test cases passed, confirming that the main functions, access controls, and email verification operated as expected.

Overall, the project achieved its main objectives by providing a centralized and organized platform for managing and accessing Life University theses. The system can help reduce the difficulties of physical or scattered thesis storage and improve access to published research materials.

5.2 Limitations of the System

Although the system achieved its main objectives, several limitations remain. The system currently supports PDF files only and does not support other formats such as DOCX. It also does not include plagiarism detection or AI-generated content detection because these functions are outside the project scope. In addition, the system focuses on completed and passed theses and does not manage the thesis writing or academic supervision process. Finally, the system was developed based on the requirements

identified during the project period, so additional requirements may arise when the system is used by more users and departments.

5.3 Future Improvements

Based on the identified limitations, several improvements are recommended for future development. The system could support additional document formats, such as DOCX, in addition the platform could be expand to publish other Life University research documents, such as department reports, journals, and other research documents with separate categories for each document type. Plagiarism detection could be added to help authorized users check thesis similarity before publication. An upload permission request feature could allow eligible students to request permission directly through the system instead of providing their information manually to the Administrator or Head of Department. The system could also be integrated with the Life University website to improve access to the thesis collection. Finally, a reporting feature could be added to provide information such as the total number of published theses and the number of theses published by each department.

Appendix A: Current Physical Storage and Condition of Thesis Documents



Figure A.1 Physical storage of thesis documents in the Computer Departmental Office

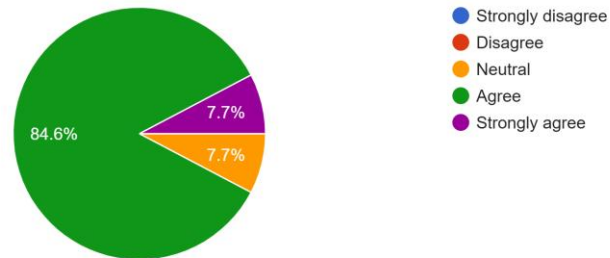


Figure A.2 Physical condition of a thesis book after being borrowed for reference

Appendix B: User Evaluation Result

1. The system is easy to use for administrators, students, and heads of departments.

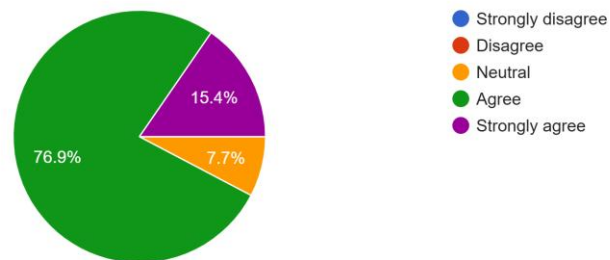
13 responses



Appendix Figure A.1 The system is easy to use for administrators, students, and heads of departments rating (Likert scale)

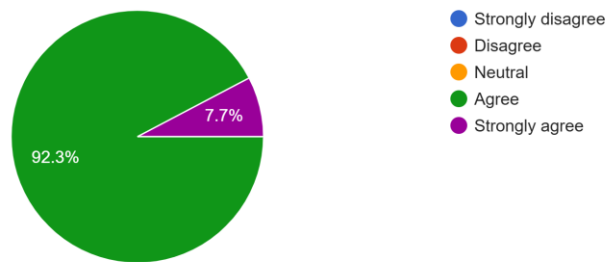
2. The system helps students and researchers find useful information easily.

13 responses



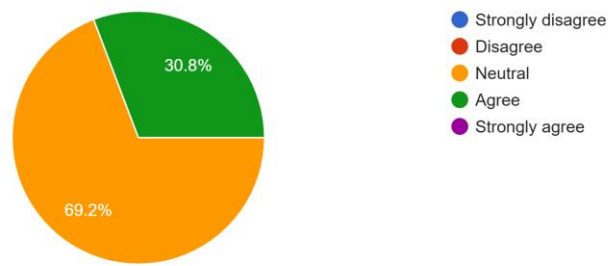
Appendix Figure A.2 The system helps students and researchers find useful information easily rating (Likert scale)

3. The search and filter functions help users find questions and answers quickly and easily.
13 responses



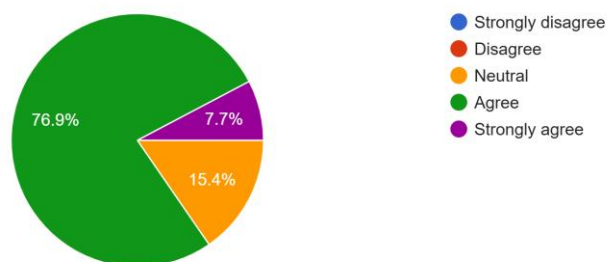
Appendix Figure A.3 The search and filter functions help users find questions and answers quickly and

4. The thesis history feature is easy to use.
13 responses



Appendix Figure A.4 The thesis history feature is easy to use rating (Likert scale)

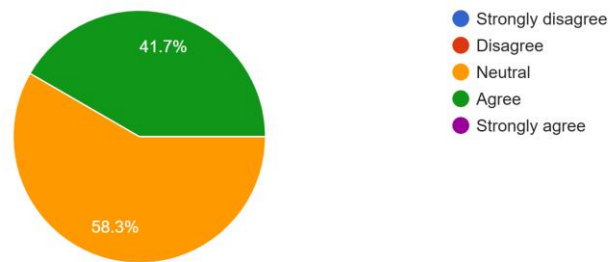
5. Viewing and downloading thesis PDF files is easy.
13 responses



Appendix Figure A.5 Viewing and downloading thesis PDF files is easy rating (Likert scale)

6. The system provides the functions needed for my role.

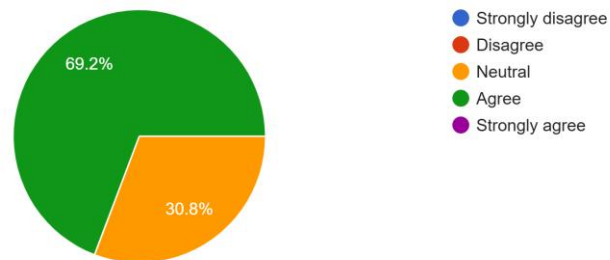
12 responses



Appendix Figure A.6 The system provides the functions needed for my role rating (Likert scale)

7. The notification feature is useful.

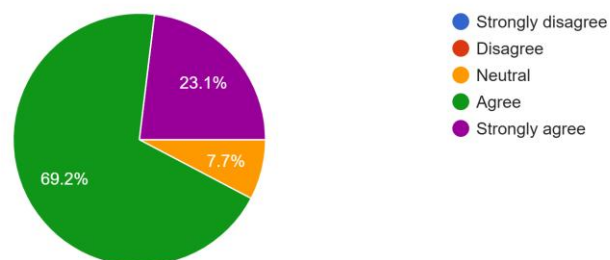
13 responses



Appendix Figure A.7 The notification feature is useful rating (Likert scale)

8. The system makes thesis management and access easier.

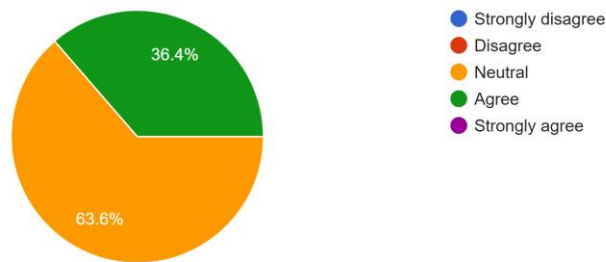
13 responses



Appendix Figure A.8 The system makes thesis management and access easier rating (Likert scale)

9. The bookmark (saved thesis) feature is useful.

11 responses



Appendix Figure A.7 The bookmark (saved thesis) feature is useful rating (Likert scale)

10. Uploading a thesis is easy.

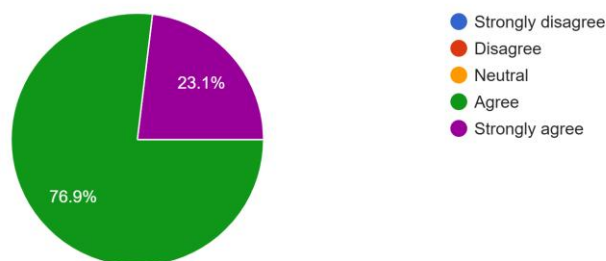
13 responses



Appendix Figure A.10 Uploading a thesis is easy rating (Likert scale)

11. The system helps save time when accessing or managing theses.

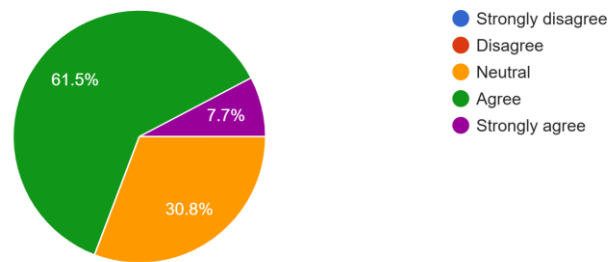
13 responses



Appendix Figure A.11 The system helps save time when accessing or managing theses rating (Likert scale)

12. The feedback provided by the Head of Department (HoD) or Administrator after rejecting thesis are useful.

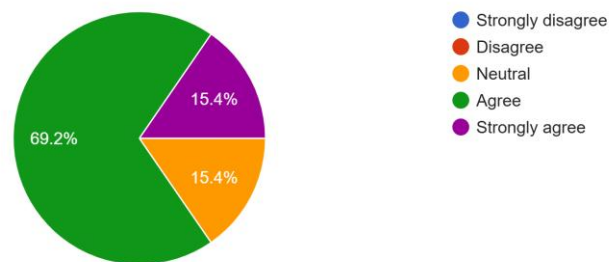
13 responses



Appendix Figure A.12 The feedback provided by the Head of Department (HoD) or Administrator after rejecting thesis are useful rating (Likert scale)

13. I am satisfied with the overall performance of the system.

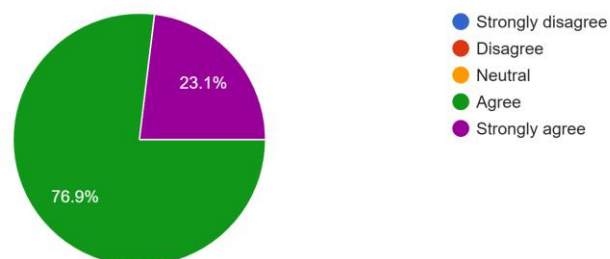
13 responses



Appendix Figure A.13 I am satisfied with the overall performance of the system rating (Likert scale)

14. I would recommend this system to other users.

13 responses



Appendix Figure A.14 I would recommend this system to other users rating (Likert scale)

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